Two spectral functionals set the exponential cost of shadow moment estimation

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Classical shadows estimate moments $\mathrm{Tr}(\rho^k)$ from randomized single-copy measurements. The finite-sample variance of the resulting U-statistic splits into projection terms of different budget order (a Hoeffding decomposition [11, 12] stated for shadows in [1], with two known decay regimes [13, 14]), and the state-dependent variances fixing their ratio have been bounded rather than evaluated. We evaluate them exactly for an arbitrary state. The exponential rate of the cost is then not a constant of the protocol but a spectral functional of the state: ζ_2 is a Pauli sum with kernel $14^{-|P|}$, giving $7^n - 1 \leq \zeta_2 \leq (17/2)^n$ for every state and confining the per-qubit base to $[7, 17/2]$ in the limit, while ζ_1 is a cubic contraction of the same second-moment identity. Their ratio is the budget $M^* = \zeta_2/(2\zeta_1)$ at which the effective error exponent crosses over, with $\mathrm{base}(M^*) = \mathrm{base}(\zeta_2)/\mathrm{base}(\zeta_1)$ exactly. The branches differ: 28/5 for spread spectra, and exactly 5 for every pure product state, where $\zeta_1 = (3/2)^n - 1$ and $\zeta_2 = (15/2)^n - 1$ at any orientation, so purity alone fixes neither factor. The law holds statewise, not merely on average: across a family whose thresholds span a decade, predicted and measured per-state error agree with rank correlation 0.87 and slope 0.97, against 0.26 on a fixed-estimand control too narrow to resolve an ordering. The threshold is also estimable from the snapshots alone, by unbiased four-block estimators whose cost grows more slowly in n than M^* and falls below it by eight qubits. Fed into two exact bias laws for collective measurement it becomes a plug-in criterion, nothing fitted to the crossover data, for when two-copy measurement attains the lower RMSE at equal copy budget, placing every resolving cell within one qubit across the five state families of the $k = 2$ held-out sweep. Results are simulation to ten qubits; at $k \geq 3$ the projection variances are numerical. The exponent migration this describes is a finite-budget effect over the sizes and budgets near-term experiments occupy: once the budget passes M^* the error returns to the $M^{-1/2}$ law, and we claim no improvement to that asymptotic scaling.

1 Introduction

At the maximally mixed state, estimating purity from classical shadows is as expensive as it ever gets. The estimand is 2^{-n}, nearly zero, and every snapshot returns $\mathrm{Tr}(G\rho) = 2^{-n}$ deterministically, so the first-order projection variance ζ_1 vanishes identically. What does not vanish, however, is the kernel term: the single-qubit second moment $\mathbb{E}[\mathrm{Tr}(G_1G_2)^2]$ equals 7 for a maximally mixed qubit, so $\zeta_2 = 7^n - 4^{-n}$, and the estimator's error is of order $7^{n/2}/M$ against a target shrinking as 2^{-n}. Low purity does not remove the exponential cost; it concentrates it.

The exponential is therefore a property of the estimator, and the question this paper answers is what fixes its rate. The answer is not the purity, and not a constant of the protocol. It is two spectral functionals of the state.

The finite-sample variance of the shadow U-statistic splits into projection terms of different budget order (the standard Hoeffding decomposition [11, 12], stated for shadows in [1]), and that the resulting decay rate changes with the budget is established: Elben et al. [13] state both regimes and plot them, and Aasen and Gärttner [14] track the exponent migrating between them under detector noise. What those analyses do not do is evaluate the coefficients. Bounding ζ_1 and ζ_2 separately constrains their ratio not at all, and the exact second moment that would fix it is applied in the literature to worst-case sample complexity rather than to a state.

Thesis. We evaluate the two projection variances exactly for an arbitrary state at $k = 2$, from the second-moment identity of [1]: ζ_2 as a Pauli sum with kernel $14^{-|P|}$, ζ_1 as a cubic contraction of the same identity. Three things follow. First, $7^n - 1 \le \zeta_2 \le (17/2)^n$ for every state, so the per-qubit exponential base is confined to $[7, 17/2]$ in the limit, and which value a state realizes is set by how its Pauli weight is distributed rather than by its purity: a Haar-random pure state and a pure product state, both of purity one, sit at 7 and at 15/2. Second, the budget at which the effective error exponent crosses over is $M^* = \zeta_2/(2\zeta_1)$, and its growth base factorizes as $\mathrm{base}(\zeta_2)/\mathrm{base}(\zeta_1)$ exactly, giving 28/5 for spread spectra and exactly 5 for every pure product state. Third, because these are statewise functionals they predict what an individual state does, which we verify by ordering states within a family whose thresholds span a decade, and (on a fixed-estimand control) show cannot be verified at all.

That the quantities are exact does not make them usable, since evaluating them needs the state. We close that gap as well, with unbiased estimators of both variances built from four disjoint blocks of a pilot shadow budget, which never form the Pauli spectrum and whose cost grows more slowly in n than the threshold they locate.

Set against two exact bias laws for collective measurement, the same machinery yields a plug-in criterion, with nothing fitted to the crossover data, for the size at which a two-copy measurement attains the lower RMSE at equal copy budget. That application is where the practical stakes sit, and they are not small: on the noisy-pure states we benchmark, at a fixed budget and ten qubits, the single-copy purity estimate projected into the interval that purity is known a priori to occupy has RMSE 0.582, while a constant fixed at the midpoint of that interval, which never looks at the data, has 0.310. From seven qubits on, the single-copy route is worse than that constant. Before projection the estimator is still unbiased; it is the spread that has become useless.

That matters because the quantity is not exotic. The moments $\mathrm{Tr}(\rho^k)$ are the entry point to much of what one wants to know about an unknown state: the second is the purity, their logarithms give the Rényi entropies, moments of the partial transpose furnish PPT entanglement criteria and bounds on negativity, and purity enters error-mitigation protocols such as virtual distillation. Any experiment characterizing the state its device actually prepared needs these numbers.

We compare two routes throughout, and propose neither: both are established strategies, and what is new here is an exact account of what each of them costs (the variance of one, the bias of the other). The single-copy route measures one copy at a time in randomized local bases, building classical shadows [1], and forms an unbiased U-statistic from the snapshots; it uses no entangling gates. The collective route holds k copies simultaneously and measures the cyclic-permutation operator, whose expectation is exactly $\mathrm{Tr}(\rho^k)$; it has $O(1)$ variance but requires entangling operations across copies. The asymptotic separation between them is settled. Single-copy strategies require exponentially more samples than collective ones for a family of learning tasks [2, 3], and for protocols restricted to an identical single-copy projection-valued measurement the lower bound for purity estimation is $\Omega(\max\{1/\varepsilon^2, 2^{n/2}/\varepsilon\})$ [4]. What is not settled is the question an experimentalist faces: at this noise level and this system size, which route has the lower RMSE at equal state-copy budget?

The gap is visible in recent work. A May 2026 study of quantum machine learning advantage on tens of noisy qubits [5] declined to give its measure-first protocol multi-copy access, citing both a result specific to its learning task and the resource demands of multi-copy entangled measurements. That task is not moment estimation, but the caution about resource cost is a premise we can test quantitatively. On the theory side, [6] shows that the exponential two-copy advantage for purity *testing* collapses under local depolarizing noise; their task is testing and ours is estimation, a distinction we take up in Section 8. The caution is reasonable. What has not been available is a prediction of the size at which it stops applying.

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2 Setting and estimators

2.1 Local random-unitary classical shadows

Let ρ be an unknown n-qubit state. A single snapshot is produced by drawing an independent Haar-random single-qubit unitary U_q for each qubit, applying $\bigotimes_q U_q$, measuring in the computational basis to obtain a bitstring b, and forming

$$G = \bigotimes_{q=1}^{n} \left(3\, U_q^\dagger |b_q\rangle\langle b_q| U_q - \mathbb{I}\right).$$

The snapshot is an unbiased estimator of the state, $\mathbb{E}[G] = \rho$. A budget of M snapshots consumes M copies of ρ.

2.2 The estimator must be the exact U-statistic

The k-th moment is estimated by the U-statistic over distinct k-tuples of snapshots,

$$U_M = \binom{M}{k}^{-1} \sum_{i_1<\cdots<i_k} h(G_{i_1}, \ldots, G_{i_k}), \qquad h(g_1, \ldots, g_k) = \frac{1}{k!} \sum_{\pi \in S_k} \operatorname{Re}\operatorname{Tr}\bigl(g_{\pi(1)} \cdots g_{\pi(k)}\bigr),$$

which is exactly unbiased: $\mathbb{E}[U_M] = \operatorname{Tr}(\rho^k)$.

The tuples are formed exhaustively rather than sampled. Forming them from already-collected snapshots is classical post-processing and consumes no additional copies, so subsampling saves nothing physical while inflating the variance: in our sweeps it inflated the single-copy RMSE by a factor of about five at $n = 2$ and more than fiftyfold at $n = 4$, enough to reverse the benchmark's conclusion. Any single-copy-versus-collective comparison must use the exact estimator on the single-copy side.

For $k = 2$ the full U-statistic has an exact closed form, and the two ways of evaluating it trade M against n. The power-sum reduction $\operatorname{Tr}(S^2) - \sum_i \operatorname{Tr}(G_i^2)$, with $S = \sum_i G_i$, is linear in M but builds dense $2^n \times 2^n$ matrices at $O(M4^n)$; we instead use the per-qubit factorization $\operatorname{Tr}(G_i G_j) = \prod_q \operatorname{Tr}(G_i^q G_j^q)$, forming an $M \times M$ pairwise-trace matrix in $O(M^2 n)$ time. At $k = 4$ no complete power-sum reduction exists, because the snapshots do not commute; Appendix A gives an exact construction over the 15 set partitions of the four cyclic slots, matching brute-force enumeration to $\lesssim 10^{-13}$.

2.3 The collective route

Let C_k be the cyclic permutation operator on k registers of dimension $d = 2^n$. It satisfies

$$\operatorname{Tr}\bigl(C_k\, \rho^{\otimes k}\bigr) = \operatorname{Tr}(\rho^k), \qquad \operatorname{Tr}(C_k) = d,$$

the second identity because the cyclic permutation has a single cycle, and it is what makes the depolarizing bias law of Section 6.1 come out linear in the noise rate.

We use the destructive SWAP test rather than the ancilla-based controlled-SWAP construction, which decomposes into many two-qubit gates and would be dominated by gate noise on current hardware. Two copies are prepared on $2n$ qubits; for each pair $(i, i+n)$ a CNOT is applied with control i and target $i+n$, followed by a Hadamard on qubit i, and all $2n$ qubits are measured. The purity is recovered by the parity rule

$$\operatorname{Tr}(\rho^2) = \sum_{\text{bitstrings}} (-1)^{\#\{\, i \,:\, a_i = b_i = 1\,\}}\, P(\text{bitstring}),$$

where a_i, b_i are the outcomes on the two copies of qubit i. The circuit uses exactly n two-qubit gates, requires no ancilla, and has depth two after state preparation; we verified the sign rule against exact simulation to 10^{-16} (Appendix B). That gate count is what keeps the collective route's measurement overhead linear in n, with no ancilla and no routing on a matched topology, a compilation fact we return to in Section 2.6.

This construction is specific to $k = 2$, where C_2 is Hermitian with eigenvalues ± 1 so its expectation is directly a two-valued measurement. For $k \geq 3$ the cyclic operator is not Hermitian and no analogous destructive parity test exists, so we take the standard ancilla-based Hadamard test: an ancilla in $|+\rangle$, a controlled-C_k on the k copies, and an X-basis measurement of the ancilla, whose ± 1 outcome has expectation $\mathrm{Re}\,\mathrm{Tr}(C_k \rho^{\otimes k}) = \mathrm{Tr}(\rho^k)$. Its outcome is bounded in $[-1, 1]$ with shot variance $1 - \mu^2$, writing μ for the expectation the test returns, and that is the model used for all k in Section 7. We call this $k \geq 3$ idealization (a bounded outcome with that shot variance, carrying no gate, depth or routing cost) the *bounded-outcome test*. The cost is not comparable: Section 2.6 gives the transpiled gate-count ratios and the ancilla and routing this circuit needs.

2.4 Copy-fair accounting

A single-copy protocol at budget M consumes M copies of ρ. A k-copy collective test at budget M consumes $\lfloor M/k \rfloor$ measurements, each using k copies (the swept budgets are not all multiples of k). All comparisons in this paper hold the total number of state copies fixed.

2.5 The state ensemble is not a free choice

The cost of shadow-based purity estimation depends on the state: at a single qubit the second moment grows with the purity, as Section 3.3 derives exactly, while at n qubits what sets the rate is the Pauli weight profile rather than the purity (Section 3.5). The consequence is a trap, though not the one it first appears to be. A random Ginibre state under heavy depolarizing noise has purity approaching 2^{-n}, and a quantity that is approximately zero is easy to estimate at fixed absolute error, though not at fixed relative precision, nor for testing, where the gap to be resolved shrinks too (Section 8). What does not follow is that the estimator is well behaved there.

Section 1's maximally mixed case is the extreme, and it is worth pricing: the two projection variances ζ_1, ζ_2 and the budget threshold M^* are defined in Section 3, and at $\rho = \mathbb{I}/2^n$ the first vanishes exactly, so $M^* = \zeta_2/(2\zeta_1)$ diverges and the estimator sits in the higher-order regime at every budget, while $\zeta_2 = 7^n - 4^{-n}$ leaves the exact variance $2\zeta_2/M(M-1)$, so the error is of order $7^{n/2}/M$, exponential in n, against a target itself shrinking as 2^{-n}.

That makes the exponential a property of the estimator, not of high-purity states. Its leading exponential rate is set by ζ_2: purity alone does not fix it, and what does is how the state's Pauli weight is distributed. That rate is 7 per qubit at the maximally mixed state and for the spread-spectrum ensembles evaluated here, and larger for stabilizer-structured spectra (Section 3.5). Purity governs ζ_1, and so M^* and the exponent, not the exponential itself. The ensemble nonetheless decides what the benchmark measures: at $\rho = \mathbb{I}/2^n$ the collective bias also vanishes identically for every *unital* channel, since $\mathrm{Tr}(\rho^k)$ then sits exactly on the depolarizing floor $2^{n(1-k)}$, so both routes degenerate at once and the comparison stops being informative. The restriction matters: amplitude damping is not unital, and carries $\mathbb{I}/2$ to a state of purity $(1 + g^2)/2$, writing g throughout for a channel noise rate as in Section 6.

We therefore benchmark on noisy-pure states, a prepared pure state under modest depolarizing noise,

$$\rho = (1-q)|\psi\rangle\langle\psi| + q\,\mathbb{I}/2^n, \qquad |\psi\rangle \sim \text{Haar}, \qquad q \in [0.05,\, 0.3],$$

whose purity is stable across the sizes tested but set by q, approaching $(1-q)^2$ at large n: about 0.90 at $q = 0.05$, 0.81 at the representative $q = 0.1$, and 0.49 at $q = 0.3$. These states keep the problem hard and meaningful at once: the moment is of order one, so an absolute error is a statement about something, and the collective route has a bias to pay, so the routes are in competition. Choosing the wrong ensemble does not make the estimator cheap; it makes the comparison vacuous.

2.6 What the collective circuit costs to compile

The collective route's overhead is often quoted as the reason not to use it. The counts below separate what is a compilation fact from what is a device measurement: they are what a transpiler returns for a stated circuit against the coupling map committed with this repository, and they need no device access to reproduce.

Transpiling the $k = 2$ destructive test of Section 2.3 onto the square-lattice map of a 108-qubit device (retrieved from the provider's API and committed as package data, with a native $\{\mathrm{CZ}, R_x, R_z\}$ basis), the $n = 2$ Bell instance uses four CZ gates (two to prepare each copy and two for the pairing) with zero routing overhead, and the GHZ ladders at $n = 3, 4$ map with zero routing at $3n - 2$ CZ gates and nest inside one another. Structured states on a matched topology are therefore cheap to pair; generic states are not, and how much they cost depends on the state-preparation synthesis, so we make no claim there.

For $k \geq 3$ the relevant circuit is instead the ancilla-based Hadamard test of Section 2.3, and it is not comparable: transpiled to the same map, the controlled cyclic shift on kn qubits uses about ten times the two-qubit gate count of the $k = 2$ destructive test at $k = 3$ and about sixteen at $k = 4$, and needs an ancilla and routing the destructive test does not. Every $k \geq 3$ result in this paper is a simulation of the idealized bounded-outcome test of Section 2.3, which carries none of that cost, and the crossover sizes Section 7 reports for $k \geq 3$ should be read as an upper bound on what the collective route buys.

These are gate counts, not error budgets. What they establish is that the entangling overhead is not the binding constraint for the structured $k = 2$ circuits we study; what binds instead is the statistical failure of the single-copy estimator, which Section 7.5 quantifies. On a real device neither gate counts nor an idealized primitive are the whole story.

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3 The projection variances

3.1 The Hoeffding decomposition gives the variance exactly

For a symmetric kernel h of order k, define the c-th order projection and its variance,

$$h_c(g_1, \dots, g_c) = \mathbb{E}_{G_{c+1},\dots,G_k}[h(g_1, \dots, g_c, G_{c+1}, \dots, G_k)], \qquad \zeta_c = \mathrm{Var}[h_c(G_1, \dots, G_c)].$$

The variance of the U-statistic is then exactly

$$\boxed{\mathrm{Var}(U_M) = \binom{M}{k}^{-1} \sum_{c=1}^{k} \binom{k}{c}\binom{M-k}{k-c}\zeta_c}$$

the standard Hoeffding decomposition [11, 12], for which we claim no novelty; what is new is the closed-form evaluation of ζ_1 and ζ_2 at $k = 2$ and the threshold their ratio fixes (Section 3.5). For $k = 2$ it reduces to

$$\mathrm{Var}(U_M) = \frac{4(M-2)\,\zeta_1 + 2\,\zeta_2}{M(M-1)}, \qquad \zeta_1 = \mathrm{Var}[\mathrm{Tr}(G\rho)], \quad \zeta_2 = \mathrm{Var}[\mathrm{Tr}(G_i G_j)].$$

We verified the general formula against brute-force Monte Carlo of the exact estimator at $k = 3$ and $k = 4$, on noisy-pure, GHZ, and low-rank states, with all ratios within 3%, and confirmed it reduces to the $k = 2$ form at a ratio of 1.000000. The law is exact, not asymptotic, at every budget.

Because the kernel is multilinear in the snapshots, every projection variance depends on the shadow ensemble only through its second-moment matrix, and local Haar single-qubit shadows and local Pauli-basis shadows have identical second-moment matrices, so all results of this section hold for either. One subtlety is easy to miss: for $k = 2$ the second-order projection *is* the kernel, so ζ_2 is the kernel variance, but for $k \geq 3$ it is not: ζ_2 is the two-argument projection and the kernel variance is ζ_k. Substituting one for the other is wrong by roughly a factor of seven and corrupts any higher-moment variance estimate while appearing to work at $k = 2$.

3.2 Two terms compete, and their ratio is a budget threshold

The exact variance contains two competing terms. At large M the linear term dominates,

$$\mathrm{Var}(U_M) \approx \frac{4\zeta_1}{M} \quad \Longrightarrow \quad \mathrm{RMSE} \propto M^{-1/2}, \qquad \alpha = \tfrac{1}{2},$$

the familiar statistical scaling a fixed-exponent bound predicts; at small M the higher-order term dominates,

$$\mathrm{Var}(U_M) \approx \frac{2\zeta_2}{M^2} \quad \Longrightarrow \quad \mathrm{RMSE} \propto M^{-1}, \qquad \alpha = 1,$$

and the two are equal at

$$\boxed{M^* = \frac{\zeta_2}{2\zeta_1}}$$

the balance point of the two large-M forms. The exact numerator terms balance at a point differing from this by an additive constant, immaterial once M^* is large, and nothing that follows depends on the choice because the predicted exponents are computed from the exact variance formula rather than from M^*. Which side of M^* an experiment sits on determines the character of its scaling, and the projection variances that set M^* are state-dependent. To see that dependence we compute them exactly for one qubit.

3.3 The single-qubit second moment, exactly

For a single qubit with density matrix r (we write r for a single-qubit state, reserving ρ for the full n-qubit state, since the identity holds only at one qubit) with Bloch length t and purity $p = (1 + t^2)/2$,

$$\boxed{\mathbb{E}\big[\mathrm{Tr}(Gr)^2\big] = \tfrac{1}{4} + \tfrac{5}{4}t^2 = \tfrac{5}{2}p - 1}$$

and the single-qubit first projection variance follows as

$$\zeta_1 = \tfrac{3}{4}t^2 - \tfrac{1}{4}t^4,$$

verified numerically across the full Bloch range. Two consequences follow. The second moment grows by a factor of six from a maximally mixed qubit, where it equals $1/4$, to a pure qubit, where it equals $3/2$; compounded over n qubits that factor is the origin of the exponential cost, and it is state-dependent rather than universal. The compounding picture is a heuristic for the factorized case, and it does not survive as a functional form: Section 3.4 shows that a weight-only quadratic ansatz for ζ_1 fails already at one qubit, before entanglement is in play. The identity also passes the boundary check: at $t = 0$ it gives $\zeta_1 = 0$, correct because $\mathrm{Tr}(Gr) = 1/2$ deterministically for a maximally mixed qubit.

3.4 The weight-only quadratic ansatz fails already at one qubit

The single-qubit result invites a generalization: if the second moment carries a factor of 3 per qubit in a Pauli string's support, then ζ_1 ought to decompose over Pauli strings with a coefficient depending only on the weight,

$$\zeta_1 \overset{?}{=} \sum_P c_{|P|} \langle P \rangle^2.$$

This provably fails at one qubit. With universal weight coefficients the most general one-qubit form is $c_0 + c_1 t^2$, affine in t^2, because the three squared Pauli expectations sum to t^2 and carry no further invariant. The exact $\zeta_1 = \frac{3}{4}t^2 - \frac{1}{4}t^4$ of Section 3.3 has a quartic term whose coefficient $-\frac{1}{4}$ no choice of c_0, c_1 reproduces. The obstruction is a quartic dependence on the state that a form quadratic in the Pauli expectations cannot carry, present before any question of entanglement arises.

The insufficiency grows with size. Against a diverse family of 19 states at $n = 2$, varying depolarizing rate, rank, and structure, the best-fit weight-only ansatz leaves a residual of up to 12.3%. The ansatz nonetheless *appears* to hold within a narrow single-parameter family, noisy-pure states at fixed $q = 0.1$ (the derivation ensemble itself), where it fits to 0.1%, because that family spans a low-dimensional invariant subspace: a closed form validated on one ensemble is not a closed form. What the weight-only form actually is, and what it omits, is the subject of the next subsection: it is the diagonal of an exact cubic identity already in [1], and the residual it leaves is that identity's off-diagonal sum.

3.5 Both variances in closed form, for any state

Section 3.4 rules out a weight-only quadratic form for ζ_1. What it does not rule out, and what holds, is a cubic one. The ingredient is already in the literature. Lemma 4 of [1] gives the exact second moment of two reconstructed Pauli coefficients under local shadows,

$$\mathbb{E}[\hat{x}_P\, \hat{x}_Q] = 3^{\,|\mathrm{supp}(P) \cap \mathrm{supp}(Q)|}\ \mathrm{Tr}(\rho\, P \ominus Q)\,,$$

when P and Q agree on every qubit where both are non-identity and zero otherwise, where $\hat{x}_P = \mathrm{Tr}(GP)$ and $P \ominus Q$ cancels the shared non-identity factors. We claim no novelty for this identity; what follows is its evaluation. Hayashi [30] restates it as a full covariance matrix over Pauli pairs, crediting the same lemma, and extends it to biased basis probabilities; that work bounds the finite-sample error of the covariance itself rather than contracting it, which is what we do below.

Since $\mathrm{Tr}(G\rho) = 2^{-n} \sum_s \langle P_s \rangle \hat{x}_s$, the first projection variance follows by contraction, and since the k-argument kernel of Section 3.1 is multilinear in the snapshots, every ζ_c is a quadratic functional of the same second-moment matrix. Two scope notes precede the evaluation.

Written out, the contraction of the second-moment identity with the Pauli expansion of ρ is, for any state,

$$\zeta_1 = 4^{-n} \sum_{P \sim Q} 3^{\,|\mathrm{supp}(P) \cap \mathrm{supp}(Q)|}\ \langle P \rangle\, \langle Q \rangle\, \langle P \ominus Q \rangle \ -\ \mathrm{Tr}(\rho^2)^2,$$

where $P \sim Q$ means P and Q agree on every qubit where both are non-identity, and $P \ominus Q$ is the string left after the shared non-identity factors cancel. Two of the three Pauli expectations multiply whenever $P \neq Q$, so the expression is cubic in the $\langle P \rangle$, which is why the weight-only quadratic form of Section 3.4 cannot match it. Concretely XI and XZ are compatible and leave IZ, while XI and YI are incompatible and contribute nothing. The diagonal $P = Q$ is the weight-only form; the off-diagonal $P \neq Q$ is the cubic remainder it drops.

As Section 3.1 noted, the projection variances do not distinguish local Haar single-qubit shadows from local Pauli-basis shadows: the two ensembles have entrywise identical second-moment matrices, so ζ_c agrees for every c and every k. Every statement below holds for either.

Everything in this subsection concerns $k = 2$. The higher moment orders are treated numerically as before; their projections do not reduce the same way, and we say what is known about them at the end.

The second projection variance is a spectral sum. The two-copy kernel is diagonal in the Pauli basis, and collecting the compatible pairs by the string they leave behind gives, for any state,

$$\zeta_2 \ =\ 7^{\,n} \sum_{P} \langle P \rangle^2 \left(\tfrac{1}{14}\right)^{|P|} \ -\ \mathrm{Tr}(\rho^2)^2,$$

the sum running over all 4^n Pauli strings with $|P|$ the weight, the kernel following from the per-qubit weights of $\mathrm{Tr}(G_1 G_2)^2$ counted in Appendix A. Only the identity term is state-independent, so the per-qubit growth rate of ζ_2 (its base, defined precisely under *The threshold's base factorizes* below) is never below 7, and a counting argument over compatible pairs bounds it above by 17/2. At the maximally mixed state every other term vanishes and the expression returns $7^n - 4^{-n}$, recovering Section 2.5. What sets the base is how the state's Pauli weight is distributed: a spectrum spread over all 4^n strings leaves the identity term dominant and the base at exactly 7, while a spectrum concentrated on a stabilizer group lifts it.

Evaluation for the noisy-pure ensemble. Averaging over the ensemble of Section 2.5, $\rho = (1-q)|\psi\rangle\langle\psi| + qI/2^n$ with $|\psi\rangle$ Haar, define the ensemble-averaged projection variances $\bar{\zeta}_c(n,q) = \mathbb{E}_\psi[\zeta_c(\rho_\psi)]$. We write the bar only for these ensemble averages; ζ_1 and ζ_2 without it denote the statewise quantities of Section 3.1. Their evaluation requires only the second and third Haar moments of Pauli expectations, $\mathbb{E}[\langle P\rangle^2] = 1/(d+1)$ and $\mathbb{E}[\langle P_u\rangle\langle P_s\rangle\langle P_{s'}\rangle] = 2/((d+1)(d+2))$ when $P_s P_{s'} = P_u$, together with four counts over compatible Pauli pairs, two for each variance. Writing $d = 2^n$, $u = 1-q$, and $p_2 = \mathrm{Tr}(\rho^2) = u^2 + q(2-q)/d$,

$$\bar{\zeta}_1 = 4^{-n}\left[1 + \frac{u^2(10^n-1)}{d+1} + \frac{2u^2(4^n-1)}{d+1} + \frac{2u^3(16^n - 10^n - 2\cdot 4^n + 2)}{(d+1)(d+2)}\right] - p_2^2,$$

$$\bar{\zeta}_2 = 4^{-n}\left[28^n + \frac{u^2(34^n - 28^n)}{d+1}\right] - p_2^2,$$

the second being the spectral sum above evaluated on this ensemble. Each qubit contributes one identity of weight 3^0 and three Paulis of weight 3^1, so $\sum_s 3^{|s|} = 10^n$; the compatible pairs give 16^n with that weighting and 34^n with its square, with diagonals 10^n and 28^n; Appendix A carries out the Haar average block by block.

The threshold reported below is the ratio of these averaged variances, $\bar{\zeta}_2/(2\bar{\zeta}_1)$, which is the quantity the state-aggregated RMSE of Section 7.2 measures; it is not the ensemble mean of the statewise ratio, and the two differ in general.

The threshold for this ensemble. For the ensemble of Section 2.5 the closed forms give $\beta \equiv \mathrm{base}(\zeta_1)$, here $\mathrm{base}(\bar{\zeta}_1) = 5/4$, and $\mathrm{base}(\bar{\zeta}_2) = 7$, so

$$M^*(n) \;\longrightarrow\; \frac{1}{2(1-q)^2}\left(\frac{28}{5}\right)^n, \qquad \frac{28}{5} = 5.6,$$

with base and prefactor both exact. The blocks below state the general structure this is one instance of, and the other branches it does not cover.

The fitted base drifts with the fitting window, from 5.15 over $n = 2$ to 7, through 5.34 over $n = 2$ to 9, to 5.60 over $n = 4$ to 9. The closed forms reproduce that drift exactly, returning 5.18, 5.36 and 5.60 over the same three windows with no sampling anywhere, so the drift is a finite-size property of the fit and 28/5 is what it drifts toward.

Two bounds, and what they do and do not attain. The spectral sum is a sum of non-negative terms whose $P = \mathbb{I}$ term is 7^n exactly, and $\mathrm{Tr}(\rho^2)^2 \le 1$; termwise $\langle P\rangle^2 \le 1$ and $\sum_P 14^{-|P|} = (1+3/14)^n = (17/14)^n$. Hence, for *every* n-qubit state,

$$\boxed{7^n - 1 \;\le\; \zeta_2(\rho) \;\le\; \left(\tfrac{17}{2}\right)^n}$$

with no condition beyond positivity and unit trace. Both are one-line consequences of the identity, and both are loose in a way worth stating. The lower bound is not attained: the infimum over states is $7^n - 4^{-n}$, reached only at $\rho = \mathbb{I}/2^n$, so the stated form is slack by $1 - 4^{-n}$; across a zoo of 70 state-size pairs the closest approach is the maximally mixed state at $n = 5$, with $\zeta_2/7^n = 0.999999942$. The upper bound is not attained and not approached: saturating it would require $\langle P\rangle^2 = 1$ for all 4^n strings at once, and projected gradient ascent over pure states reaches only 76.5% of $(17/2)^n$, its maximiser at $n = 2$ being the pure product state. We report it as the ceiling a counting argument gives, not as a value states approach.

What follows from the lower bound is a statement about the *limit*, and the distinction is not pedantic. Since $\zeta_2 \ge 7^n - 1$, any state family for which $\mathrm{base}(\zeta_2)$ exists has $\mathrm{base}(\zeta_2) \ge 7$. The finite-$n$ ratio $\zeta_2^{1/n}$ is *not* bounded below by 7: the subtracted $\mathrm{Tr}(\rho^2)^2$ is $O(1)$ against 7^n, and twelve of those 70 records fall below, as low as 6.982. Seven of the twelve are at $n = 2$, where the subtraction bites hardest; the remainder are the maximally mixed and near-mixed states, which sit just under 7 at every size. The per-qubit base is a limit, and only the limit obeys the bound.

The threshold's base factorizes. For a sequence a_n write $\mathrm{base}(a) = \lim_{n\to\infty} a_n^{1/n}$ when the limit exists. Then, for any state family,

$$\boxed{\mathrm{base}(M^*) \;=\; \frac{\mathrm{base}(\zeta_2)}{\mathrm{base}(\zeta_1)}}$$

whenever both limits exist *and* both leading coefficients are nonzero. The second condition does real work rather than decorating the first: at $\rho = \mathbb{I}/2^n$ we have $\zeta_1 = 0$ exactly, M^* is infinite, and $\mathrm{base}(M^*)$ is undefined, while $\mathrm{base}(\zeta_2) = 7$ is perfectly well behaved. Where the conditions hold the identity is exact, and we have checked it on eight families whose bases genuinely differ (noisy-pure, Haar-pure, pure product, GHZ, a linear graph state, two product families and rank-2 Ginibre), recovering $\mathrm{base}(M^*)$ from the ratio to within 8×10^{-16}.

Both factors are spectral functionals of the state rather than universal constants, and that is the substance of the result: the exponential cost of shadow moment estimation has a rate the state chooses, through how its Pauli weight is distributed, within a window the identity fixes.

The branches, and one family that is exact throughout. The same two functionals evaluated on the families this paper uses do not agree:

state family	β	$\mathrm{base}(\zeta_2)$	M^* base	status
noisy-pure ($q = 0.1$)	$5/4$	7	$28/5 = 5.60$	exact, Haar-averaged
Haar-pure	$5/4$	7	$28/5 = 5.60$	exact, the $q = 0$ case
low-rank (rank 2)	$\to 5/4$	7	5.60	numerical; see below
pure product, GHZ	$3/2$	$15/2$	5	exact at every n

The last row is exact in a stronger sense than the spectral sum alone gives. For a single pure qubit $\sum_P \langle P\rangle^2 14^{-|P|} = 1 + t^2/14 = 15/14$ whatever the Bloch direction, and $\mathbb{E}[\mathrm{Tr}(Gr)^2] = 3/2$ likewise; both factorize over qubits for a product state. So for *every* pure product state, at any orientation,

$$\zeta_1 = \left(\tfrac{3}{2}\right)^n - 1, \qquad \zeta_2 = \left(\tfrac{15}{2}\right)^n - 1, \qquad M^* = \frac{(15/2)^n - 1}{2\left[(3/2)^n - 1\right]},$$

verified against the general evaluator on random orientations to 10^{-14} and 10^{-11} respectively. The whole family therefore has an exact $M^*(n)$ with base exactly 5, and the GHZ state, whose stabilizer group splits into even-weight Z strings and weight-n strings, gives $\zeta_2 = \frac{1}{2}[(15/2)^n + (13/2)^n] - \frac{1}{2}$ and the same base.

Purity does not determine either factor. A Haar-random pure state and a pure product state both have $\mathrm{Tr}(\rho^2) = 1$ and sit on different branches, at $28/5$ and at 5. What separates them is the weight profile the kernel $14^{-|P|}$ reads off, which is why $28/5$ is the spread-spectrum value and not a constant of the protocol. The low-rank row is the one numerical entry: $\beta \to 5/4$ asymptotically like the other spread spectra but reads ≈ 1.21 at the sizes evaluated here, and its M^* base uses the asymptotic value.

Every base quoted from a finite window in this paper is a fit, not a limit, and we label it so. The drift is real and one-sided: for pure product states, where $\beta = 3/2$ exactly, a log-linear fit of ζ_1 returns 1.666 over $n = 3$ to 5 and 1.540 over $n = 6$ to 8, approaching $3/2$ from above.

Higher moment orders. The multilinearity argument above shows every ζ_c depends only on the second-moment matrix, so closed forms exist in principle at $k \geq 3$ as well. We did not obtain them: at $k = 3$ and 4 the state-dependent contributions dominate the state-independent ones, so the kernel bases are not read off the maximally mixed values, and the required counts over compatible tuples are beyond what we derived. The $k \geq 3$ coefficients used below are estimated numerically.

3.6 Evaluating them for a given state

The identities of Section 3.5 hold for any state, so they can be evaluated for one. What that costs, and what it needs as input, decides whether the threshold is a usable quantity or only a structural one.

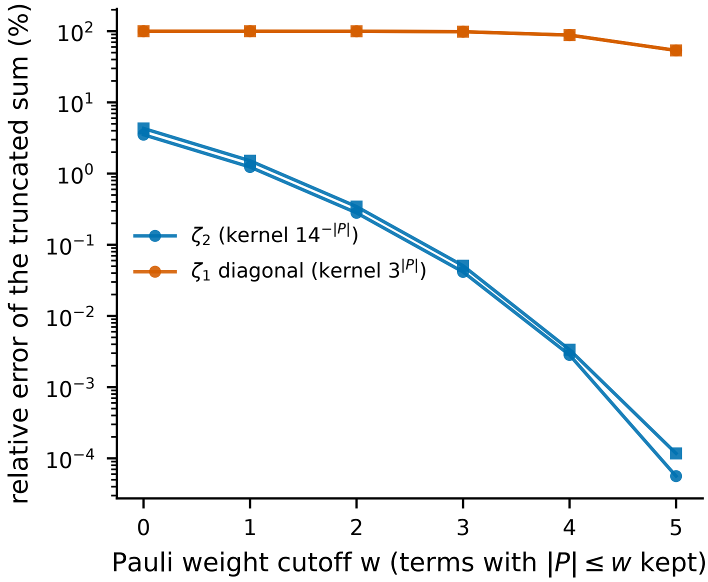

Figure 1: The two projection variances read opposite ends of the Pauli spectrum. Relative error of each functional when its sum over Pauli strings is truncated to weight $|P| \leq w$, at $n = 6$ (circles noisy-pure, squares Haar-pure). ζ_2 carries the kernel $14^{-|P|}$, which suppresses high weight, so a few percent of the strings determine it. The ζ_1 diagonal carries $3^{|P|}$, which amplifies high weight, so most of the spectrum is needed. The cutoff $w = n$ is exact by construction and is not plotted.

Given the 4^n Pauli expectations, ζ_2 is a single weighted sum over strings and costs $\Theta(4^n)$. The cubic ζ_1 runs over compatible *pairs*, and per qubit exactly ten local patterns are compatible: $(\mathbb{I}, \mathbb{I})$, $(\mathbb{I}, a)$, $(a, \mathbb{I})$ and (a, a) for the three non-identity letters, so the pair set has size 10^n and the cost is $\Theta(10^n)$, which is $N^{\log 10/\log 4} = N^{1.661}$ in the input size $N = 4^n$. Enumerating a tail of qubits vectorized while looping over a head keeps peak memory at $O(10^{\text{tail}})$ independently of n. Measured on one core, ζ_1 takes 2.3 s at $n = 9$, 23.5 s at $n = 10$ and 242 s at $n = 11$, so $n = 10$ is the largest size evaluable in under a minute; ζ_2 is milliseconds throughout. The evaluator is exact in the sense that no sampling enters: it reproduces $\zeta_1 = \frac{3}{4}t^2 - \frac{1}{4}t^4$ at one qubit to 10^{-12}, the maximally mixed and pure-product values to machine precision, and the Haar average of Section 3.5 to within the sampling error of the average itself.

The input is the whole Pauli spectrum, which is the state, and no proper subset suffices, but the two functionals fail differently, and the asymmetry is the useful part. Their kernels pull in opposite directions on Pauli weight. Truncating the sums to $|P| \leq w$ at $n = 6$ on the noisy-pure ensemble, ζ_2 is recovered to 1.2% from weight one and to 0.3% from weight two, which is 154 of the 4096 strings, because its kernel $14^{-|P|}$ suppresses high weight. The ζ_1 diagonal carries $3^{|P|}$, which amplifies it: keeping 82% of the strings still leaves 54% error, so the top weight shell alone carries more than half (Fig. 1). High-weight expectations are exactly the ones local shadows estimate worst, their variance growing as $3^{|P|}$, so ζ_1 is the hard factor and ζ_2 the easy one. Section 5 shows that this does not block estimation from data, because the estimator there never forms the spectrum at all.

3.7 The α transition

Fitting the sampled projection variances over the windows below gives clean exponential scalings, retained here for the contrast with the closed forms of Section 3.5, which return 5.18 for the M^* base over $n = 2$ to 7 against the regression's 5.15. Over that window,

$$\zeta_1 \approx 0.63 \cdot (1.35)^n, \qquad \zeta_2 \approx 1.10 \cdot (6.93)^n, \qquad M^* \approx 0.87 \cdot (5.15)^n,$$

and over $n = 2$ to 9, $M^* \approx 0.76 \cdot (5.345)^n$; the ζ_1 base is mildly curved, about 1.35 at small n and 1.30 over the full range, which accounts for the difference. The kernel variance grows roughly five times faster per qubit than the linear variance, so their ratio M^* grows approximately exponentially over the range we fit. The consequence is direct: at fixed budget, increasing n pushes M below M^*, the higher-order term takes over, and the effective exponent α, the negative slope of $\log \mathrm{RMSE}$ against $\log M$, fitted over the budget range used (Appendix C), so window-averaged and shifting with n, migrates from 1/2 toward 1, and past it for higher k:

n	$M^*(n)$	predicted α	measured α
2	≈ 25.6	0.501	0.495 ± 0.013
4	≈ 549	0.528	0.528 ± 0.012
9	$\approx 3.0 \times 10^6$	0.998	1.006 ± 0.023

The $M^*(n)$ column is the exact threshold evaluated from the closed-form projection variances, not the fitted law evaluated at n. The predictions are computed from the projection variances (closed-form at second order, estimated from shadow snapshots at higher) fed into the exact variance formula, with nothing fitted to the α data (Fig. 2): the derived law matches 8 of 8 within two standard errors at $k = 2$, 6 of 7 at $k = 3$ with the single miss on the two-sigma boundary and seed-sensitive, and 5 of 5 at $k = 4$.

The exact combinatorial structure is necessary, though $k = 2$ is the wrong place to see it. There the large-M truncation of Section 3.2, $\mathrm{Var} \approx 4\zeta_1/M + 2\zeta_2/M^2$, tracks the exact variance to within 0.05% across the budgets used and reproduces the measured α at 8 of 8, because at $k = 2$ the second projection and the full kernel variance coincide and the truncation discards nothing. They separate at higher orders: keeping the same two terms with their correct coefficients, $k^2\zeta_1/M + \frac{1}{2}k^2(k-1)^2\zeta_2/M^2$, gives 15 of 20 across $k = 2, 3, 4$ against the exact law's 19 of 20, and misses $k = 3$, $n = 8$ by twenty-six standard errors: by then ζ_3/ζ_2 exceeds 10^4 and the discarded terms carry the variance.

A log-linear regression of $M^*(n)$ over $n = 2$ to 9 returns base 5.34, 95% CI [5.14, 5.54], prefactor 0.75 ± 0.09, and ζ_1, ζ_2 fit to bases 1.30 $[1.26, 1.34]$ and 6.94 $[6.90, 6.98]$; the ζ_2 base sits just below the exact per-qubit 7 at the maximally mixed state (Section 2.5) and barely moves across windows, the drift carried by ζ_1, in the manner Section 3.5 anticipates. Appendix C gives the residual and leave-one-out detail.

A fixed-exponent bound $\mathrm{RMSE} \leq C_n/\sqrt{M}$ is not wrong, and the tighter two-term treatments do not assume a fixed exponent either [13]. What it describes is the wrong regime for the systems being run. At nine qubits and accessible budgets the estimator sits near $\alpha = 1$, where extra shots reduce error faster than $1/\sqrt{M}$ but from a variance already exponentially large, so any extrapolation presuming $\alpha = 1/2$ errs predictably. At fixed n the linear term dominates once the budget grows without bound, so the asymptotic scaling remains the square-root law: the migration reported here is a finite-budget effect over the window hardware reaches, not a change in asymptotic sample complexity.

Having characterized the single-copy cost exactly, we turn to what it predicts for an individual state.

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4 Statewise validation

Section 3 makes a statewise claim: $\zeta_1(\rho)$ and $\zeta_2(\rho)$ are functionals of a particular state, so they should predict what *that* state does, not only what its family does on average. Testing that is harder than it looks, and most of this section is about the obstacle rather than the result.

4.1 What a statewise test requires

A cell of states can discriminate a statewise law from a family-average law only if the law's predictions actually differ across the states in it, by more than the noise on the measurement they are compared against. Write s for the relative spread of the predicted RMSE across a cell and η for the relative noise on each measured RMSE; the cell is informative only when $s/\eta > 1$. On

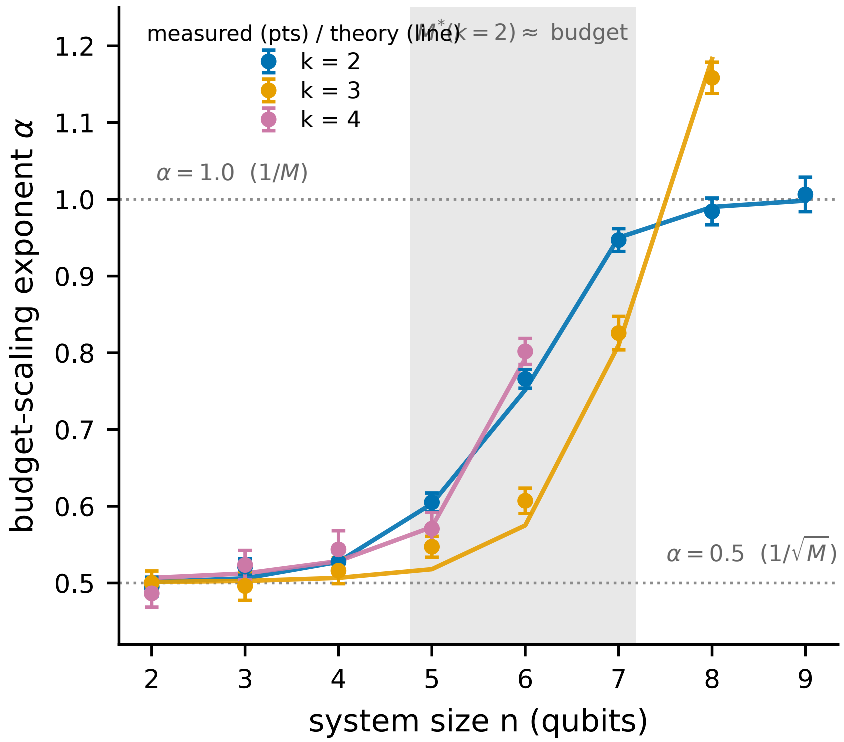

Figure 2: The effective budget-scaling exponent transition, the paper's central phenomenon. Measured budget-scaling exponent α (RMSE $\propto M^{-\alpha}$, fitted over budgets 2000–128000; markers with bootstrap standard errors) versus system size n for moment orders $k = 2, 3, 4$, with the derived-law prediction (lines). *The prediction curves are computed from projection variances fed into the exact variance formula of Sec. 3.1, with no parameter fitted to the plotted points.* As n grows at fixed budget and M falls below the threshold M^*, α migrates continuously from $1/2$ toward 1, and past it for the higher moment orders.

the three families the crossover sweeps hold out, it is not, as the top three rows below show; the fourth, noisy-pure, is the derivation family, has no row here, and is quantified as the control in Section 4.2. The last two rows are the families added below to fix this.

state family	estimand	M^* max/min	s	η	s/η
Haar-pure	fixed	1.20	2.4%	5.9%	0.40
low-rank (rank 2)	6.0%	1.42	3.5%	5.9%	0.59
GHZ-noisy	fixed	1.00	0	11.8%	0
variable-q	21.6%	2.54	8.8%	5.9%	1.49
variable-rank	73.8%	9.74	27.0%	5.9%	4.58

All rows are at $n = 4$; s is computed from the exact statewise variances and η from the trial count of the sweep.

The reason is structural, not a matter of effort. Haar-pure has an estimand fixed by construction (as does the noisy-pure control, since $\mathrm{Tr}(\rho^k)$ is a function of (n, q, k) alone and does not depend on $|\psi\rangle$), and GHZ-noisy is a single state, so its spread is exactly zero. Low-rank does vary, by 6.0% in the estimand, but its extreme statewise thresholds differ by a factor of only 1.42. No affordable trial count rescues any of the three: with the extremes that close together there is almost nothing to order.

We therefore added two families whose parameters are drawn per state rather than fixed: variable-q, which is the noisy-pure construction with $q \sim U[0.05, 0.45]$ drawn for each state, and variable-rank, a Ginibre state of rank drawn uniformly from 1 to 8. Both cost exactly what their fixed-parameter counterparts cost to sample. Variable-rank spans nearly a factor of ten in statewise M^* and three quarters of its mean in the estimand itself, which is what makes a statewise test possible at all.

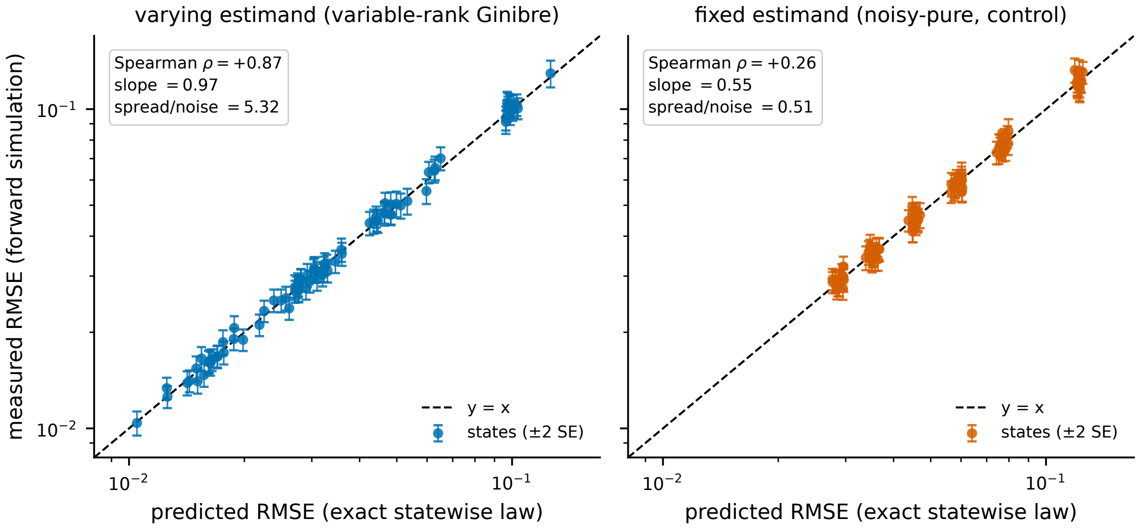

Figure 3: Statewise validation. Predicted versus measured single-copy RMSE for individual states at $k = 2$, $n = 3, 4, 5$ and two budgets, 14 states per cell, 300 trials each. *Predictions come from each state's exact* ζ_1, ζ_2, *with nothing fitted to the plotted points.* Left: a varying-estimand family, where the statewise threshold spans an order of magnitude, so the law is tested state by state. Right: the fixed-estimand noisy-pure family as a negative control; its statewise spread is below the measurement noise, so no ordering can be resolved and a family-average law would score the same. Both panels share one scale.

4.2 The law orders individual states

On variable-rank at $n = 3, 4, 5$, two budgets, 14 states per cell and 300 forward-simulation trials per state (so $\eta \approx 4.1\%$ against a predicted spread of 21.9%, a spread-to-noise ratio of 5.3 rather than the 4.58 tabulated above at $n = 4$ alone, because this run pools three sizes and buys more trials per state), the exact statewise law reproduces each state's RMSE with a median relative deviation of 2.7%, and 82 of 84 state-budget pairs agree within two standard errors. The ordering is recovered, not just the scale: the rank correlation between predicted and measured RMSE within a cell averages $\rho = +0.87$ with a minimum of $+0.49$, and regressing measured on predicted gives a slope of 0.97, a one-to-one relation across the decade the family spans (Fig. 3, left).

The same run on noisy-pure is the control, and it behaves as the diagnostic says it must: the median deviation is comparable at 2.4% and 79 of 84 pairs agree within two standard errors, but $\rho = +0.26$ and the slope is 0.55, because with $s/\eta = 0.51$ there is no ordering to recover (Fig. 3, right). A law that read only the family mean would score the same there. Reporting the control alongside is the point: it is what distinguishes a statewise result from an average one, and it is why the earlier sweeps, run entirely on fixed-estimand families, could not have established this.

Two weaker statewise checks run over a wider grid, six families and $n = 2$ to 6 at three budgets with 24 trials each: there the exact law predicts per-state RMSE to a median 9.4%, consistent with the 14% trial noise of that run, with 562 of 615 state-budget points inside two standard errors, and the per-state budget-scaling exponent within two standard errors in 194 of 205 cases. A third check changes the unit. Carrying one seed across sizes to form state *sequences*, since a single state lives at one size and cannot cross over by itself, the crossover criterion of Section 7.1 places the crossover exactly in 82 of the 123 sequence-channel cells this construction yields (a different population from the 123 swept family cells of Appendix C, which happens to be the same size), and within one qubit in all 123. The exact rate is lower than the family-aggregate figures of Section 7.2, which is the honest cost of asking a sharper question.

4.3 The crossover sweep on the new families

Adding the two families to the held-out crossover sweep extends it from three ensembles to five, 45 swept cells at $k = 2$ over three channels and three rates. Three checks fix the comparison to what was already committed: the measured crossover on the original three ensembles reproduces the earlier sweep in all 27 cells for the unbiased estimator and all 27 for the range-projected one (the same estimator clipped to the interval $[d^{1-k}, 1]$ that every moment must lie in, defined in Section 7.3), and substituting the exact statewise projection variances for the sampled ones on the prediction side moves the predicted crossover in none of the 27. The exact evaluator is therefore free to adopt: it changes no number the earlier sweeps reported.

Over all five families the criterion resolves 27 crossovers and places 26 exactly and all 27 within one qubit for the unbiased estimator, and 26 resolving with 21 exact and all 26 within one for the range-projected estimator; scoring the non-crossing cells as predictions as well gives 42 of 45 under either estimator. On the two new families alone the unbiased figures are 12 resolving, all 12 exact, and 18 of 18 once non-crossing cells are counted.

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5 Estimating the threshold from a pilot budget

The threshold is exact but its inputs are the state, and an experimenter has snapshots. This section closes that gap: M^* can be estimated from the snapshots themselves, without ever forming the Pauli spectrum whose high-weight part Section 3.6 shows is both necessary to ζ_1 and badly estimated by shadows. The route around that is to estimate the two variances *as variances* rather than through their spectral representations.

5.1 Four blocks and two unbiased estimates

Split a pilot of M snapshots into four disjoint blocks A, B, C, D and write $K_{ij} = \mathrm{Tr}(G_i G_j) = \prod_q \mathrm{Tr}(G_i^q G_j^q)$. Then

- $\zeta_2 = \mathrm{Var}[K_{ij}]$ is estimated by the sample variance of K over the disjoint pairs (A_i, B_i). Each pair is independent, so this is unbiased directly.
- $\mathbb{E}[\mathrm{Tr}(G\rho)^2]$ is estimated by $\mathrm{mean}_{i\in A}\, \bar{K}_{iB}\, \bar{K}_{iC}$ with $\bar{K}_{iB} = \mathrm{mean}_{j\in B} K_{ij}$. Because B and C are disjoint from each other and from A, the two inner means are independent unbiased estimates of $\mathrm{Tr}(G_i\rho)$, so their product is unbiased for $\mathrm{Tr}(G_i\rho)^2$.
- $\mathrm{Tr}(\rho^2)^2$ is estimated by $\bar{K}_{AB}\, \bar{K}_{CD}$ over the four disjoint blocks, unbiased because the factors are independent and each is unbiased for $\mathrm{Tr}(\rho^2)$.
- ζ_1 is the difference of the last two.

Nothing here is a plug-in approximation; each piece is exactly unbiased, which is what lets a small pilot be believed. Every block mean the construction needs is a mean of the 4^n-dimensional vector $x_i = \bigotimes_q a_i^q$ of per-qubit Pauli coefficients, since $\mathrm{Tr}(G_i G_j) = 2^n \langle x_i, x_j \rangle$ for Hermitian factors, so the whole computation is $O(M 4^n)$ with no pairwise matrix ever formed.

The structure also says which factor will be hard. ζ_1 is a difference of two estimates sharing the scale $\mathrm{Tr}(\rho^2)^2$, each carrying noise of order ζ_2/M; since $\bar{\zeta}_2$ grows as 7^n while $\bar{\zeta}_1$ grows as β^n with $\beta = 5/4$ on the ensemble of Section 2.5, the difference is taken between quantities whose common part dominates. That is what the measurements show: at $n = 8$ and a pilot of 32,000 the relative error of $\hat{\zeta}_2$ is 12.7% while that of $\hat{\zeta}_1$ is 205%, and $\hat{\zeta}_1$ comes out non-positive in 17 of 48 draws, in which case the ratio is undefined rather than large and we report it as such.

5.2 How much pilot it takes

Table 1 gives the median relative error of $\widehat{M^*}$ against the exact statewise M^* for the same state, on the noisy-pure family. The error falls steeply with the pilot budget, and the budget needed for a given accuracy grows with n, but much more slowly than the threshold it is locating.

The comparison that matters is against M^* itself, since a pilot larger than the threshold it locates would be self-defeating. Measured, the ratio of the 10% budget to M^* falls from ≈ 300 at $n = 2$ through 1.94 at $n = 6$ and 1.40 at $n = 7$ to 0.98 at $n = 8$: the pilot budget grows about twofold per qubit against the threshold's roughly fivefold, so the overhead becomes relatively cheaper as n grows, and it crosses unity between $n = 7$ and $n = 8$. Interpolating log-linearly

n	M^*	10% budget	ratio	error at that budget
2	26	8,000	307	5.5%
3	110	8,000	72.6	6.5%
4	541	8,000	14.8	8.8%
5	2,884	32,000	11.1	5.8%
6	16,519	32,000	1.94	9.0%
7	91,741	128,000	1.40	8.4%
8	520,935	512,000	0.98	5.7%

Table 1: Pilot cost against the threshold it locates, noisy-pure ensemble. The M^* column is the median *statewise* threshold over the states drawn here, so it differs by a few percent from the ensemble-averaged $M^*(n)$ tabulated in Section 3.7. The 10% budget is the smallest *swept* budget whose median relative error on $\widehat{M^*}$ fell below 10%, so it is an upper bound set by the granularity of the grid and not an interpolation; the last column gives the error actually achieved there. Full error-versus-budget curves are in Fig. 4. The ratio crosses unity between $n = 7$ and $n = 8$.

between those two bracketing measured points puts the crossing just below $n = 8$, at $n \approx 7.95$ (Fig. 4). We state this as measured rather than extrapolated because an earlier fit over $n \leq 6$ alone put it near $n = 7$ while understating the required budget by a factor of 1.85 at $n = 7$ and 3.8 at $n = 8$; the crossing survives that correction only because M^* grows faster than the pilot does.

The tail of that convergence was worth a second look, because it is where an unbiased estimator can still mislead. At sixteen repetitions the last octave at $n = 8$ appeared to flatten, falling only from 8.4% to 7.3%, which is what an error floor would look like. It is not one. Re-running the three largest budgets on independent states at three states by forty repetitions (119 to 120 usable draws each) gives 27.7%, 10.9% and 5.7%, a fitted exponent of -1.14 with a 95% confidence interval of $[-1.38, -0.88]$. That is at least as steep as the $M^{-1/2}$ an unbiased variance estimate should show, and it excludes the flattening: the earlier appearance was an artifact of the repetition count. Pooling the same draws, the median of $\widehat{M^*}$ relative to M^* is 0.992, 0.972 and 1.004 across the three budgets, scattering about unity rather than settling at an offset, which is the other signature a floor would leave and does not.

The higher-precision run does move one number, and we report the conservative reading. Its median deviation at 256,000 is 10.9% rather than the 8.4% the sixteen-repetition run gave, so that budget now sits just above the 10% line and the bracket at $n = 8$ becomes 512,000. Two independent determinations straddle the threshold there, which is precisely the grid-granularity fragility Table 1 is caveated for; taking the larger moves the ratio at $n = 8$ from 0.49 to 0.98 and the crossing from $n \approx 7.3$ to $n \approx 7.95$. Table 1 and Fig. 4 use the larger value. The qualitative claim is unchanged (the pilot cost falls below the threshold it locates by eight qubits), and we prefer the version that claims less. A bracket read off a doubling grid is a coarse instrument near the threshold, and should be read as one.

The varying-estimand family is harder in absolute terms, and we report it separately rather than pooled, because pooling would hide the reason. On variable-rank the 10% budget is 8,000 at $n = 2$, 32,000 at $n = 4$, 128,000 at $n = 6$, and more than 256,000 at $n = 7$, where the error is still 19.7% at that budget. But its thresholds are correspondingly larger (a median M^* of 503,444 at $n = 7$ against 91,741 for noisy-pure), so the two move together: rank-deficient states have smaller ζ_1, which raises M^* and makes ζ_1 harder to estimate at once. The ratio, not the budget, is the quantity that transfers.

What this does not provide is a performance guarantee. The estimators are unbiased and their convergence is measured, but bounding the probability that a pilot puts an experiment on the wrong side of M^* needs a concentration bound for $\hat{\zeta}_1$, which is a difference of degree-3 and degree-4 statistics in snapshots whose kernel variance grows as 7^n; the fourth moment that a Bernstein-type argument would need is itself a higher-order projection variance we do not have in closed form. We state the convergence we measured and leave the guarantee open.

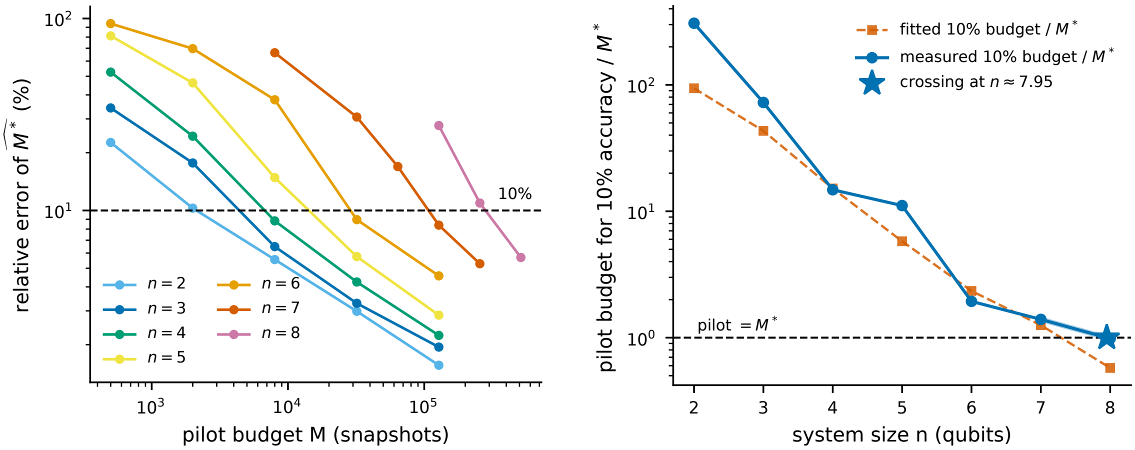

Figure 4: Estimating the threshold from data. *Left:* median relative error of $\widehat{M^*}$ against the exact statewise M^* versus pilot budget, noisy-pure, one curve per size; the dashed line marks 10% accuracy. *Right:* the 10% budget relative to M^* itself. Circles are the smallest swept budget that achieved 10%, so they are upper bounds set by the grid; the first three sit at the same raw budget of $8{,}000$ snapshots, which is why the ratio falls so steeply across them. Squares are a log-log fit through the usable range. The star marks where the measured ratio crosses unity, interpolated between the two measured sizes that bracket it. Below the dashed line the pilot costs less than the threshold it locates.

6 The collective route: two exact bias laws

At nonzero noise, once the shot budget is large enough that the finite-shot term has decayed, the collective route's error is dominated by a bias rather than a variance; at zero noise or small budgets the finite-shot variance dominates instead. The bias is independent of the shot budget, a prediction Section 7.4 tests directly and which holds. Its functional form is fixed by the geometry of the noise channel, its magnitude by geometry and rate together, and two channel classes give two exact laws.

6.1 Global depolarizing noise: linear in g, with no compounding

Let the kn-qubit collective register be subject to global depolarizing noise at rate g, so that $\rho^{\otimes k} \mapsto (1-g)\rho^{\otimes k} + g\,\mathbb{I}/d^k$ with $d = 2^n$. Then

$$\mathrm{Tr}\big(C_k\left[(1-g)\rho^{\otimes k} + g\,\tfrac{\mathbb{I}}{d^k}\right]\big) = (1-g)\,\mathrm{Tr}(\rho^k) + g\,\frac{\mathrm{Tr}(C_k)}{d^k} = (1-g)\,\mathrm{Tr}(\rho^k) + g\,2^{\,n(1-k)},$$

using $\mathrm{Tr}(C_k) = d$ from Section 2.3. Hence

$$\boxed{\mathrm{bias} = g\left|\mathrm{Tr}(\rho^k) - 2^{\,n(1-k)}\right|}$$

At a fixed g the bias is linear in g and does not compound across qubits: the single-cycle property of C_k collapses the compounding that a per-qubit form $1-(1-g)^{kn}$ would predict, and which overestimates the bias by a factor of five to fourteen over the rates swept in Section 7. The two forms are parametrically different in n, the compounding form saturating toward unity while the true bias stays linear. This holds at fixed g; on hardware the effective global rate may itself grow with register size, depth, or gate count.

6.2 Per-qubit channels: the noise relabels the state

Let $\mathcal{E}$ be any single-qubit channel borne by the collective route and not by the single-copy route, applied to every qubit of every copy held for the test, with ρ the reference state and $\mathrm{Tr}(\rho^k)$ the estimand. That $\mathcal{E}$ is specific to this route is a modeling choice rather than a derivation, and it is what makes the quantity below a bias.

The identity needs one further precondition, and it is narrow: the k copies must be independent and must undergo $\mathcal{E}^{\otimes n}$ *before* an otherwise ideal cyclic-permutation measurement. It does not cover noise inside the cyclic test (imperfect controlled-permutation gates, ancilla decoherence, readout error), nor correlated noise acting across copies, nor any channel acting during or after the circuit. Those perturb the measured observable or the instrument rather than the state being measured, and the relabeling does not apply to them. Within that precondition,

$$\mathcal{E}^{\otimes nk}\big(\rho^{\otimes k}\big) = \big(\mathcal{E}^{\otimes n}(\rho)\big)^{\otimes k} = \sigma^{\otimes k}, \qquad \sigma \equiv \mathcal{E}^{\otimes n}(\rho),$$

and the cyclic test of a product of identical states returns the k-th moment of that state:

$$\boxed{\text{measured} = \mathrm{Tr}(\sigma^k), \qquad \sigma = \mathcal{E}^{\otimes n}(\rho)}$$

so the bias is $\left|\mathrm{Tr}(\sigma^k) - \mathrm{Tr}(\rho^k)\right|$. The noise does not corrupt the measurement in the usual sense; it relabels which state is measured, and that substitution is the entire bias. A channel common to both routes would bias the single-copy route by the same substitution and cancel from the comparison, carrying no information about which route has the lower RMSE at equal state-copy budget; the law requires that $\mathcal{E}$ be borne by the collective route alone. Preparation noise is not excluded but already absorbed into ρ by the ensemble of Section 2.5, which both routes see and which contributes no bias.

6.3 Verification

Both laws are exact identities rather than approximations, and we verified them accordingly: against explicit construction of C_k and the noise-damaged k-copy state, at $n = 2, 3$ and $k = 2, 3, 4$, for global depolarizing, amplitude damping, and dephasing, agreeing to approximately 10^{-15}. We take the single-qubit Kraus operators at rate g in the standard form: amplitude damping as $K_0 = \mathrm{diag}(1, \sqrt{1-g})$ and $K_1 = \sqrt{g}\,|0\rangle\langle 1|$; dephasing as $K_0 = \sqrt{1 - g/2}\,\mathbb{I}$ and $K_1 = \sqrt{g/2}\,Z$; and global depolarizing in the closed form of Section 6.1. The first two are applied to every qubit of every copy; the third acts on the whole kn-qubit register, which is how Section 6.1 defines it. They continue to hold at three times the rates the crossover sweeps of Section 7.2 use, and on state ensembles they had never been tested against: Haar-pure ($q = 0$); low-rank, $\rho = GG^\dagger$ with G a $2^n \times 2$ complex Ginibre matrix normalized to unit trace, whose spectrum is not the rank-one-plus-identity form the noisy-pure closed forms assume; and GHZ-noisy, $(1-q)|\mathrm{GHZ}\rangle\langle\mathrm{GHZ}| + q\,\mathbb{I}/2^n$ at $q = 0.15$ with $|\mathrm{GHZ}\rangle = (|0\cdots 0\rangle + |1\cdots 1\rangle)/\sqrt{2}$. An exact identity is not expected to degrade under extrapolation, and it does not.

With the single-copy variance in closed form at $k = 2$ and the collective bias an exact identity, the crossover follows by setting one against the other.

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7 The crossover

7.1 The law

The single-copy error of the unbiased estimator grows exponentially in n and falls with the budget; Section 7.3 takes up the range-projected alternative, for which it instead saturates. The collective error is a bias floor plus a finite-shot variance: the floor is bounded above by $1 - d^{1-k}$ with $d = 2^n$, hence by unity, and is independent of the budget, while the shot term vanishes as the budget grows. The crossover size n^* is where the single-copy error rises above the collective error,

$$\underbrace{\sqrt{\mathrm{Var}(U_M)}}_{\text{exponential in } n\text{, falls with } M} = \underbrace{\sqrt{\mathrm{bias}(g,k,n)^2 + \frac{1-\mu^2}{N}}}_{\text{bounded floor + shot noise}}$$

Here μ is the noisy collective signal the cyclic test returns and $N = \lfloor M/k \rfloor$ the number of k-copy measurements a copy budget of M allows; the shot term is the variance of that bounded outcome, leaving the bias floor as the large-budget limit. Both sides come from Sections 3 and 6, evaluated on the state's projection variances and the noise channel, with nothing fitted to the data being predicted.

The bounded-floor claim is about *estimation* at fixed additive precision ε, where the bounded collective outcome costs $O(1/\varepsilon^2)$ shots at every n and the error saturates at the floor rather than growing. It is not a claim that collective measurement is cheap for all tasks: where the target precision must itself shrink exponentially in n, exponentially many samples are required even with two copies [6], and Section 8 gives the quantitative reconciliation.

7.2 Validation on 83 cells and four ensembles

Across 83 cells spanning moment order $k \in \{2, 3, 4\}$, three noise models (depolarizing, amplitude damping, dephasing), noise rates from 0 to 0.3, sizes $n = 2$ to 9, four budget multipliers, and four state ensembles, the criterion predicts 82 of the 83 simulated crossovers within one qubit and 73 of 83 exactly (Figs. 5 and 6). Those figures are for the unbiased U-statistic of Section 3, the estimator the variance law describes. For the range-projected estimator of Section 7.3, 11 cells stop resolving a crossover and the criterion places all 72 that remain within one qubit, 63 of them exactly. Both are reported throughout: they describe two different estimators, and neither supersedes the other.

Neither aggregate is homogeneous, and the split matters for reading it. The 83 are 68 noisy-pure cells whose measured n^* comes from the paired state-level z-test against measured collective data, plus 15 held-out cells whose measured n^* comes from the measured single-copy RMSE against the theory collective floor; for the range-projected estimator the 72 are 58 and 14 of the same two kinds. Appendix C states the three rules and which produced which figures, and defines the censoring and rescoring conventions the counts here and in Section 4 both use.

The comparison is asymmetric in two deliberate respects. Evidentially, the single-copy side of each cell is a forward simulation (fresh snapshots reduced by the U-statistic, so the variance law is tested against data it did not generate), while the collective side is the exact bias law evaluated analytically plus its shot noise. In the noise model, the stronger assumption, the channel of Section 6 degrades the collective route only, the single-copy route being the ideal noiseless baseline, on the reading that a joint k-copy measurement is the harder thing to implement cleanly. Real shadow readout is not noiseless, so this understates the single-copy cost, in the direction that costs us: simulating readout on both routes, with the collective parity corrected exactly through its outcome-dependent per-pair contraction rather than a blanket factor, moves the crossover to smaller n in nine of the ten cells that resolve under both models and leaves the tenth unchanged. The magnitude depends on device-specific per-qubit rates, which are committed with the repository, but the direction is what matters here, so the sizes reported below are a conservative upper bound. Under uncorrected readout the single-copy U-statistic acquires a bias that does not shrink with the copy budget, so both routes then carry error floors and the comparison changes character; the noise-rate dependence of n^* is by construction a property of the collective side.

These cells demonstrate consistency with the derived identities: three of the four ensembles (Haar-pure, low-rank, GHZ-noisy) were never used in deriving the law (Fig. 7), so they test whether the plug-in architecture stays accurate when its projection-variance inputs are re-estimated outside the derivation ensemble. What transfers is the structure of the formula, not the noisy-pure coefficients, so this is not zero-shot prediction from universal constants. GHZ states are maximally non-Haar-typical and were included because an ensemble-tuned theory ought to fail on them; the theory's accuracy there is within a factor of about 1.7 of its accuracy elsewhere, the same order of magnitude rather than a breakdown. None of the four can carry a statewise test, for the reason Section 4.1 sets out, and Section 4.3 extends the sweep to two varying-estimand families and ties it back to the counts here.

7.3 Physical constraints and the choice of estimator

Every moment obeys $\mathrm{Tr}(\rho^k) \in [d^{1-k}, 1]$ with $d = 2^n$, a range fixed before any data is taken. An estimate falling outside it can therefore be replaced by its Euclidean projection onto that interval,

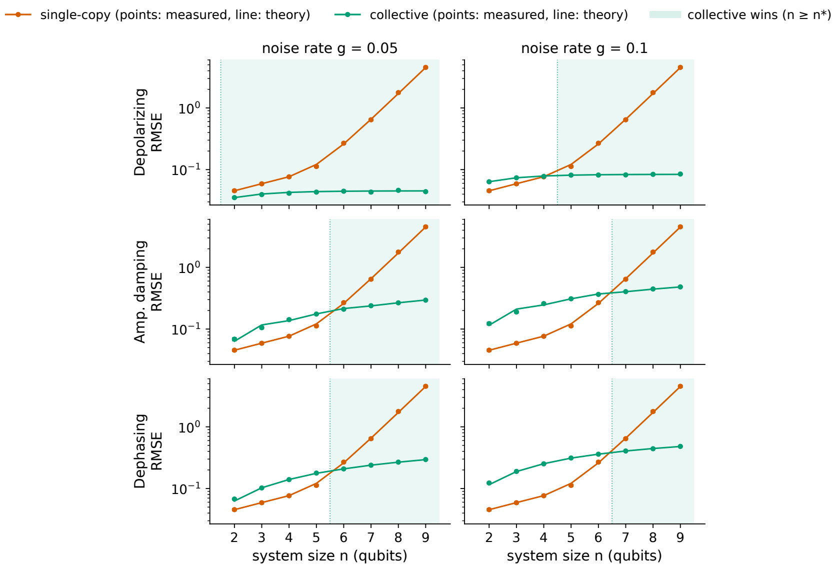

Figure 5: The crossover for purity ($k = 2$). Single-copy classical-shadow RMSE (vermillion) and collective two-copy RMSE (green) versus system size n at a fixed copy budget, for three noise channels. *The theory curves are predictions with no parameters fitted to these points, from the exact variance and bias laws (Secs. 3 and 6).* The crossover n^* is where the exponentially growing single-copy error overtakes the collective error, a bounded bias floor plus a finite-shot variance that vanishes as the budget grows.

$$\widehat{U}_M^{\mathrm{c}} = \min\bigl\{\max\{\widehat{U}_M,\, d^{1-k}\},\, 1\bigr\},$$

a contraction onto a convex set containing the truth, which cannot increase the squared error of any individual estimate, pointwise, with no distributional assumption. Over 4320 samples spanning $n = 2$ to 10 it increased it zero times.

The effect is not marginal. On the headline configuration at $n = 10$, 97.9% of raw estimates fall outside the physical range, and projection cuts the RMSE from 11.98 to 0.58, a factor of 20.6.

Section 3 is untouched: it gives the variance of the unbiased U-statistic, and that remains accurate; the closed-form Hoeffding prediction is 11.92 at $n = 10$ against 11.976 measured. What changes is which estimator an experimenter would use, and the criterion has to be re-derived for it.

Doing so and re-running the full sweep, the criterion predicts the crossover size within one qubit in all 72 cells that resolve a crossover (100.0%), exactly in 63 of those 72 (87.5%), and correctly on 119 of the 123 swept cells (96.7%) once non-crossing cells are scored as predictions, against 82 of 83, 73 of 83, and 118 of 123 for the unbiased estimator. The criterion is more accurate under projection on two of the three measures and marginally less on the third: within one qubit rises from 98.8% to 100.0% and the all-cells rate from 95.9% to 96.7%, while exact placement slips from 88.0% to 87.5%. Bounding both errors compresses the predicted and measured curves onto the same scale, so cells that straddled the boundary in the unbounded comparison stop straddling it.

The cost is a smaller denominator, and it falls almost entirely in one place. Projection removes the crossover entirely from 11 cells, so 72 rather than 83 resolve; the loss is 10 cells on noisy-pure (68 to 58) and one on GHZ-noisy (6 to 5). Where a crossover survives it never moves earlier: 60 of the 72 are unchanged, 10 move one qubit later, and 2 move four. Haar-pure keeps both its cells, each one qubit later; low-rank keeps all seven, unmoved.

Projection makes the single-copy estimator biased, so Section 7.1's comparison becomes one of two error floors (the same change of character the readout simulation of Section 7.2 imposes on both routes), and as there it does not reverse the predicted ordering.

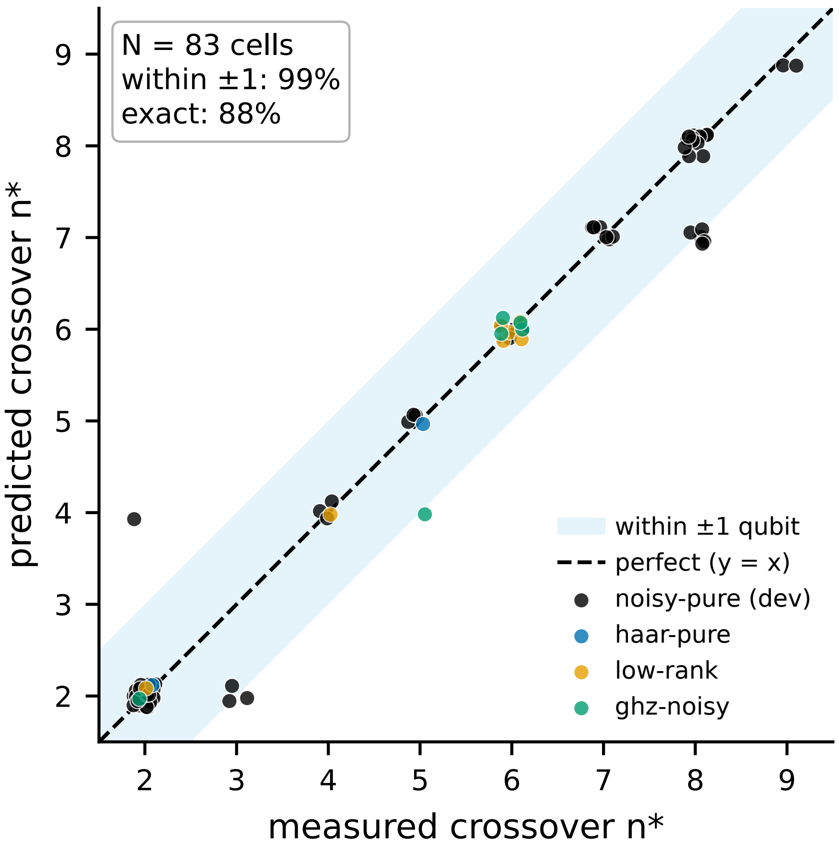

Figure 6: Crossover validation across 83 cells. Predicted versus Monte-Carlo-observed crossover size n^* over moment orders $k \in \{2, 3, 4\}$, three noise models, and four state ensembles (three unseen by the theory). Accuracy for the unbiased estimator (83 resolving cells, as 68 noisy-pure plus 15 held-out) and for the range-projected estimator (72, as 58 plus 14) is reported in Secs. 7.2 and 7.3; the two subsets are scored by different rules (Appendix C). The cells are simulation, not hardware; for the higher moment orders the collective side is the idealized bounded-outcome test of Sec. 2.3, carrying no gate, depth, or routing cost.

7.4 Three qualitative predictions, and the one we got wrong

Two of the three we predicted correctly in advance; the third we got backwards, and the law corrected us. Higher noise moves the crossover later on the channels and rates we swept, a larger bias floor taking longer to climb over; a larger budget moves it later too, the single-copy error falling with budget while the floor does not. The collective RMSE plateaus exactly at the predicted floor while the single-copy RMSE keeps falling, a direct test of the bias-versus-variance distinction the whole crossover rests on.

Higher k also moves the crossover later on the families we tested, against the naive argument that higher moments compound the single-copy variance and should cross earlier. $\mathrm{Tr}(\rho^k)$ shrinks with k, so the absolute bias floor is lower and single-copy holds out to larger n: the signal-against-a-fixed-floor argument dominates the variance-compounding one. This is an observation about the ensembles and channels swept here, not a general rule: $|\mathrm{Tr}(\sigma^k) - \mathrm{Tr}(\rho^k)|$ need not fall monotonically with k for an arbitrary spectrum and channel, and no universal attenuation rate fits the measured biases, the bias being not a rate but however much the channel deforms that particular state's spectrum, so two states of identical purity can be deformed by different amounts. Purity ($k = 2$) crosses at $n \approx 6$ to 7 under per-qubit amplitude damping and dephasing at the rates swept, and earlier under global depolarizing, where the measured crossover averages four qubits against six for the other two channels; $k = 3$ and $k = 4$ hold out to $n \approx 8$ (sizes for the idealized bounded-outcome test of Section 2.3, not for its realization, whose gate cost Section 2.6 gives).

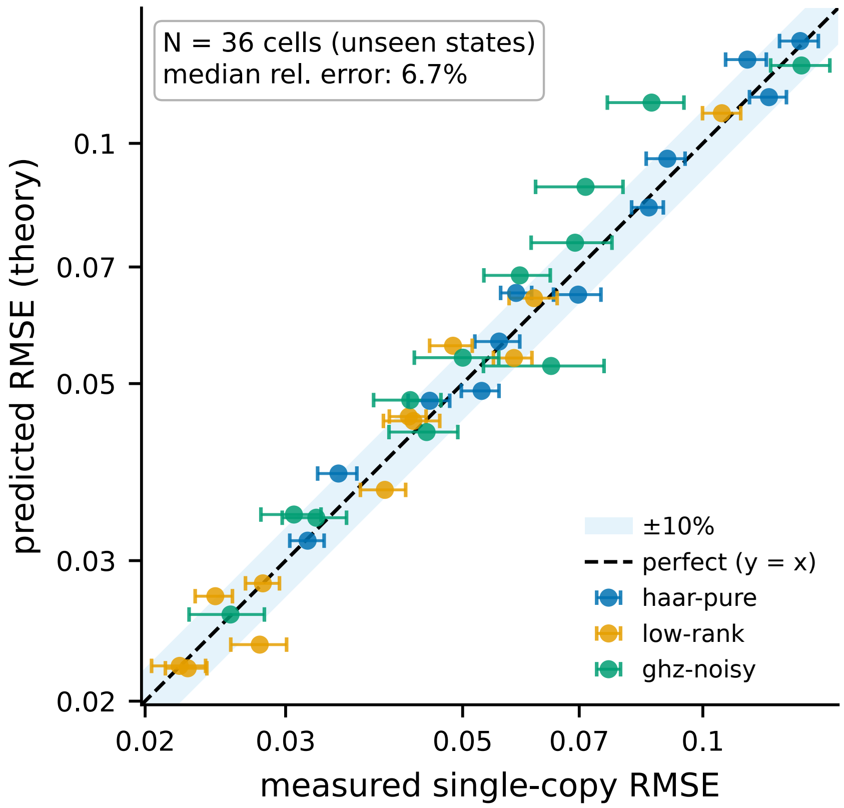

Figure 7: Out-of-ensemble validation of the variance law. Predicted versus measured single-copy RMSE on three ensembles the theory never saw (Haar-pure, low-rank, GHZ-noisy). The scatter about the diagonal is finite-trial estimation noise (median relative deviation 6.7%; Sec. 9), not a systematic error.

7.5 Where single-copy estimation stops being useful

The practical statement is a single sequence. Using the exact copy-fair unbiased estimator on noisy-pure states at a fixed copy budget, the single-copy purity RMSE runs

$$0.043 \rightarrow 0.072 \rightarrow 0.270 \rightarrow 1.62 \rightarrow 11.98$$

at $n = 2, 4, 6, 8, 10$, growing by a factor of about two per qubit on average and accelerating to roughly 2.7 per qubit over the last step, against a true purity of approximately 0.81 (Fig. 8). Projected into the physical range as in Section 7.3, the same estimates give 0.043, 0.071, 0.227, 0.500, 0.582: bounded, and saturating rather than diverging.

The bounded sequence carries the sharper statement, because it no longer rests on an unbounded scale. A constant fixed at the midpoint of $[2^{-n}, 1]$, the interval the purity is known a priori to occupy, and all that a guess which never looks at the data can use, has RMSE 0.308 at $n = 7$ against the projected single-copy estimator's 0.400, and 0.310 at $n = 10$ against its 0.582. From seven qubits on, the single-copy route is worse than that constant. The collective route holds out three qubits longer, beating the same constant through $n = 9$ and falling behind it only at $n = 10$ (0.320 against 0.310). The collective route stays bounded over the same range because its error is a bias floor: both $\mathrm{Tr}(\sigma^k)$ and $\mathrm{Tr}(\rho^k)$ lie in $[d^{1-k}, 1]$, so their difference is bounded by the width of that interval and cannot exceed $1 - d^{1-k} < 1$. What is universal is the weaker statement that the floor stays below unity, and that is what the crossover argument uses.

This speaks to the resource-cost concern behind [5]'s choice to keep its measure-first protocol single-copy. The entangling overhead that motivates such caution is real and is not the binding constraint: at the tested budgets the local-randomized-measurement shadow U-statistic we study becomes statistically unusable for moment estimation at the sizes where that overhead is being weighed, a statement about this protocol, not about every single-copy strategy.

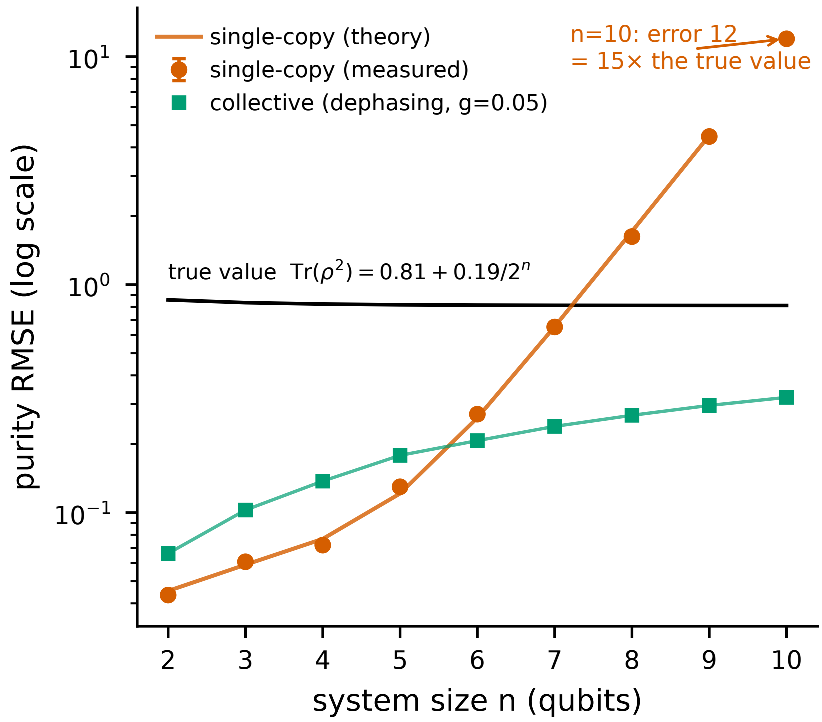

Figure 8: Single-copy error growth against a bounded collective error. Single-copy purity RMSE for the unbiased U-statistic versus n out to $n = 10$ at a fixed copy budget of $M = 2000$ (log scale; measured points with 68% CIs, theory line with no parameters fitted to the crossover data; noisy-pure ensemble, collective route under dephasing at $g = 0.05$), with the exact per-n true purity $\mathrm{Tr}(\rho^2) = 0.81 + 0.19/2^n$ marked (0.858 at $n = 2$, approaching 0.81). By $n = 10$ the single-copy error reaches ~ 12 while the collective error (green) stays bounded. Projecting the single-copy estimate into the physical range (Sec. 7.3) caps it at 0.582, but does not restore its usefulness: from $n = 7$ on it is beaten by a constant fixed at the midpoint of that range, which uses no data at all.

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8 Related work

Classical shadows and variance bounds. The classical-shadow framework [1] gives sample-complexity bounds via the shadow norm; for second-order nonlinear functionals the variance splits into a kernel term bounded by $4^{|AB|}$ and linear terms bounded by $2^{|AB|}$ [8], which states and applies the bound citing its own references, so we attribute the statement rather than its origin. These are worst-case, state-independent bounds. For local randomized measurements, Huang, Wang and Xu [15] give worst-case bounds whose exponential base depends on the measurement ensemble, for the related task of distributed inner-product estimation; these remain envelopes rather than evaluated coefficients or a threshold, as do the companion bounds for structured circuits below a 4-design [27]. The exact finite-M decomposition is the standard Hoeffding identity [11, 12], already stated as an equality for shadows by [1] and again by Elben et al. [13], who read the two-regime scaling off it. Reference [13] bounds ζ_1 and ζ_2 rather than evaluating them; [1] does supply the exact second moment, in its Lemma 4, but applies it to worst-case sample complexity rather than evaluating the coefficients for a state or an ensemble. Neither gives a threshold in M, and separate upper bounds constrain the ratio not at all, which is why the threshold does not follow from them. What this work adds is the evaluation of Section 3.5, the resulting threshold, the spectral functionals fixing its growth base, and the comparison against the collective route. Alternative shadow ensembles [9] target different observable classes. Deliberately biased shadow estimators have been studied directly: [19] rescales the estimator by a tuned factor, shrinking it toward zero, and argues the trade is always worth making in the finite-sample regime. The projection of Section 7.3 accepts bias for the same reason but by a different mechanism: it has no free parameter and no shrinkage target, and moves an estimate only when that estimate is infeasible. That an estimator reporting values on the boundary of the feasible set can itself be improved by hedging inward is known from tomography [26].

U-statistics and the Hoeffding decomposition for quantum moments. Straeter, Tsesmelis and Kwek [10] construct unbiased U-statistic estimators for the partial-transpose moments p_2 and p_3 from randomized homodyne data and derive their variance via a Hoeffding decomposition, obtaining sample-complexity bounds for a p_3-PPT entanglement criterion. This is the closest methodological neighbor to Section 3; the differences are the platform, the target (entanglement detection rather than moment estimation and the collective-measurement trade-off), and the result, since their projection variances are bounded rather than evaluated for a state and no change of scaling regime is considered. Two further analyses treat the shadow U-statistic itself. Jnane et al. [16] write the exact Lee variance for U-statistics of shadow snapshots, then bound the projection terms by a quasiprobability norm and keep only the leading $1/N$, the step that hides the second regime. Pelecanos et al. [17] study the same estimator for $\mathrm{Tr}(\sigma^k)$ under the uniform POVM and prove an explicit variance bound for it; that bound is an inequality in d for a worst-case state rather than an evaluation for an ensemble, and locates no threshold. Cioli et al. [29] write the same pairwise U-statistic for purity under shallow shadows and bound its variance with both regimes present, an M^{-1} term and an $(M-1)^{-2}$ term; the two coefficients are worst-case bounds rather than evaluations, and their ratio is never formed, so no threshold budget follows from them. The Efron–Stein decomposition, the operator-valued relative of the Hoeffding one, has separately been used in quantum estimation to quantify post-measurement damage in online shadow tomography [31], a different use of the same tool. We claim no novelty for the Hoeffding technique itself.

Crossovers, hybrids, and cheaper collective routes. A crossover in shot cost between Pauli and Clifford shadow ensembles has been reported for multipartite entanglement witnesses [7]; that is a crossover in the choice of measurement ensemble, not between single-copy and collective measurement. Zhou and Liu [18] interpolate between the routes instead of choosing, applying randomized measurements to a few coherently prepared copies; the measurement counts they report order the protocols without locating a size at which the ordering changes. A parallel line lowers the cost of the collective route itself [21, 22, 28], through Newton–Girard recursions, qubit reuse, or coherent evaluation of the functional; these improve the side our criterion compares against rather than moving where it crosses.

Sample-complexity lower bounds for purity. Gong et al. [4] improve the lower bound for purity estimation to $\Omega(\max\{1/\varepsilon^2,\ 2^{n/2}/\varepsilon\})$ for protocols using an identical single-copy projection-valued measurement. Our exact variance is consistent with these bounds and supplies the state-dependent constant they leave open, by evaluating the second moment of [1] over the ensemble. Du et al. [20] observe that single-copy purity estimation already saturates an $\Omega(\sqrt{d})$ bound and give a protocol matching it for $\mathrm{Tr}(O\rho^2)$; like the above, these are asymptotic statements in d and silent about the finite-budget coefficient. Ye, Liu and Deng [23] sharpen the picture into a hierarchy: estimating $\mathrm{Tr}(\rho^{k+1}O)$ needs $\Omega(\sqrt{d}/(k\varepsilon^{1/(1+k)}))$ samples under any k-replica protocol while $k+1$ replicas need a constant, which at $k=1$ and $O=\mathbb{1}$ says that one further copy removes the exponential cost of purity estimation. That result is worst-case over states and asymptotic in d: they establish that the separation exists, not the size at which a given experiment meets it. Two concurrent results bound the k-dependence instead, at $\Omega(k\|O\|^2/\varepsilon^2)$ samples for $\mathrm{Tr}(O\rho^k)$ [24, 25], dimension-independent and so orthogonal to the growth in n at issue here.

Noise and the collective advantage: testing versus estimation. The exponential single-copy versus collective separation is established in [2, 3]; its fate under noise is the subject of [6], the closest theoretical work to ours. For purity *testing* under per-qubit depolarizing noise $D_\lambda(\rho) = (1-\lambda)\rho + \lambda I/2$, their Theorem 2.4 proves a sample-complexity lower bound of

$$\Omega\Big(\min\Big\{\, 2^{n/2},\ \Big(\tfrac{4}{1+3(1-\lambda)^4}\Big)^n \Big\}\Big),$$

writing $b(\lambda) = 4/(1+3(1-\lambda)^4)$ for that base, and it holds against *any* algorithm, including arbitrary adaptive joint POVMs on the $2n$ qubits.

The relationship to our result turns on the task. They test, deciding between hypotheses whose gap shrinks exponentially under noise; we estimate at fixed additive ε, where the bounded collective-test outcome costs $O(1/\varepsilon^2)$ shots regardless of n. Their exponential cost comes from the testing task requiring an exponentially small ε, not from the SWAP test being intrinsically exponential for estimation. We do not evade their bound; we solve a different problem. The two results nonetheless share a mechanism: our Section 6.2 identity, that a per-qubit channel makes the k-copy test return exactly $\mathrm{Tr}(\sigma^k)$ with $\sigma = \mathcal{E}^{\otimes n}(\rho)$, is the finite-$n$ mechanism underneath their asymptotic theorem, since depolarizing drives $\mathrm{Tr}(\sigma^2)$ toward 2^{-n} under both hypotheses and the testing gap closes. Feeding our exact law into their setup at $\lambda = 0.1$, the gap $\Delta(n, \lambda) = \mathrm{Tr}(\sigma^2_{\text{pure}}) - 2^{-n}$ falls from 0.54 at $n = 2$ to 0.21 at $n = 10$, and the implied SWAP-test shot count $1/\Delta^2$ rises from 3.5 to 22 over that window. Their bound is asymptotic with its constant left open, so the comparison available is one of exponential bases rather than counts at any size: the asymptotic base of $1/\Delta^2$, $\big(4/(1+3(1-\lambda)^2)\big)^2$, equals $b(\lambda)$ at $\lambda = 0$ and exceeds it at finite noise, the difference controlled exactly by $3(1-(1-\lambda)^2)^2 \geq 0$; over $n \leq 10$ the large prefactor dominates this small-base exponential, so the growth rate measured on that window sits just below $b(\lambda)$. At our rates and sizes their bound imposes no practical obstruction, because $b \approx 1.35 < \sqrt{2}$ at $\lambda = 0.1$, so the $2^{n/2}$ branch never binds and the cost grows slowly; it becomes an obstruction only at much larger n. Their own framing agrees that the asymptotic limit at fixed noise is not the near-term regime, and that what is needed is a quantitative condition on λ for a two-copy advantage at fixed n. Our crossover criterion is that condition.

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9 Limitations

The exact evaluation is $k = 2$ only, and it is exponential. The identities of Section 3.5 hold for any state, but only at second order: at $k \geq 3$ we did not obtain them and the coefficients used are estimated numerically, so every higher-moment result here rests on sampled inputs. Evaluating them for a given state costs $\Theta(10^n)$ and takes the whole 4^n-entry Pauli spectrum as input, which caps exact evaluation near ten qubits on one core and means the exact functional is not itself a field-usable rule; Section 5 is the route that is. And the asymptotic bases and prefactors, though exact, are approached slowly: at the sizes reached here $M^*(n)/5.6^n$ is still some fifteen percent below its limit, and every base we quote from a finite fitting window is labelled a fit for that reason.

The statewise claim rests on families built to test it. None of the four families the crossover sweeps use meets the discriminating condition of Section 4, so the ordering result comes from two families we constructed for the purpose, and it is a claim about those families' range of spectra rather than a demonstration that the law orders arbitrary states. The negative control is reported alongside precisely so this is visible.

The pilot estimator has no performance guarantee. Both estimators of Section 5 are exactly unbiased and their convergence is measured, but we do not bound the probability that a pilot places an experiment on the wrong side of M^*. Doing so needs a concentration bound for a difference of degree-3 and degree-4 snapshot statistics whose kernel variance grows as 7^n, and the fourth moment it would require is a higher-order projection variance we do not have in closed form. The reported budgets are also upper bounds set by the granularity of the swept grid, not interpolations, and at small n the grid is coarse enough that they overstate the requirement considerably.

The $k = 3$ validation is 6 of 7. The single miss sits on the two-sigma boundary and is sensitive to the random seed. We report it rather than round it to 7 of 7.

The noise model is a channel abstraction. Global depolarizing and independent per-qubit channels are idealizations. The bias laws are exact given the channel, and they say nothing about whether the channel describes any particular device. The raw counts from a superconducting backend that would test this are committed with the repository and are not analysed here.

Statistical caveats. The out-of-ensemble RMSE validation carries a median relative deviation of 6.7% between predicted and measured values (Fig. 7). We investigated this and found it to be finite-trial noise on both sides rather than a systematic error: the estimator is approximately Gaussian at the budgets used, the projection variances are converged to better than 1%, and the measured RMSE converges to the predicted value to within ±0.3% as the trial count increases. The

6.7% figure matches the intrinsic RMSE noise at the trial counts used, and the signed deviation is +3.4%, which is within noise once cell correlations are accounted for. The predicted crossover n^*_{pred} is reported as an integer throughout, and we do not propagate the projection-variance estimation uncertainty onto it. Resampling the projection variances by their committed relative spread shifts the predicted crossover for 5 of the 60 noisy-pure cells (mean per-cell probability 1.7%), always by exactly one qubit and never by more.

One constrained estimator is analysed. Section 7.3 treats Euclidean projection onto $[d^{1-k}, 1]$, the constraint that follows from the range alone and needs no tuning; other constrained estimators, such as shrinkage toward the depolarizing floor, trade bias against variance differently. The crossover sizes we report for the projected estimator are specific to that choice.

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10 Conclusion

The exponential cost of estimating $\mathrm{Tr}(\rho^k)$ from local classical shadows is a property of the estimator, and its rate is a property of the state. Both statements are made precise by evaluating the two projection variances the Hoeffding decomposition already isolates: ζ_2 is a Pauli sum whose kernel $14^{-|P|}$ confines the per-qubit base to $[7, 17/2]$, ζ_1 is a cubic contraction of the same second-moment identity, and the budget threshold their ratio defines has base $\mathrm{base}(\zeta_2)/\mathrm{base}(\zeta_1)$ exactly, under a nonzero-leading-coefficient condition that the maximally mixed state violates. Neither bound on ζ_2 is attained, and the finite-size ratio $\zeta_2^{1/n}$ can sit below 7; it is the limit that the bound governs.

Two consequences are worth separating from the machinery. Purity does not determine the rate: a Haar-random pure state and a pure product state, both of purity one, sit on branches at 28/5 and at exactly 5, and for every pure product state the threshold is available in closed form at every n. And the popular value 28/5 is the spread-spectrum case rather than a constant, which is the sort of thing that is easy to over-generalize from one ensemble.

Because these are functionals of a state rather than of a family, they can be checked statewise, and we did: across a family whose thresholds span a decade the law orders individual states with rank correlation 0.87 and a slope of 0.97. The negative control matters as much as the result: on the families the crossover literature has used, including our own earlier sweeps, the statewise spread sits below the achievable measurement noise, so no such claim could have been established there at all.

The threshold is also reachable from data. Unbiased estimators built from four disjoint blocks of a pilot budget recover M^* without forming the Pauli spectrum, to ten percent from a pilot that grows about twofold per qubit against the threshold's roughly fivefold, so the overhead becomes relatively cheaper with size and falls below the threshold by eight qubits. What is missing is a performance guarantee, which needs a concentration bound for a difference of degree-3 and degree-4 statistics and is a separate problem.

Paired with two exact bias laws for the collective route, distinguished by the geometry of the noise channel rather than by its rate, the variance law predicts, with no parameter fitted to the crossover outcomes, the size at which collective measurement attains the lower RMSE at equal copy budget. Across the five state families of the $k = 2$ held-out sweep (two of them built for this paper, because the other three cannot test a statewise claim), it places every resolving cell within one qubit. For estimating purity of noisy-pure states under per-qubit amplitude-damping or dephasing noise at the rates and budgets tested here, single-copy classical shadows stop being the lower-RMSE route beyond roughly six to seven qubits; the higher moments hold out to about eight for the idealized bounded-outcome test, whose $k \geq 3$ realization costs about an order of magnitude more two-qubit gates (Section 2.6), and under global depolarizing noise the crossover falls earlier still.

Three questions follow from our work. Whether the higher-order projection variances admit the same treatment, which would extend the exact account past $k = 2$ and is the obstacle to a performance guarantee. Whether the same variance law, applied to the partial-transpose moments, gives an exact sample-complexity account of entanglement negativity estimation, which is the application that motivated us. And whether the branch structure (base 7 for spread spectra, 15/2 for stabilizer-structured ones) has a classification behind it, since the two values we can prove exactly are the two extremes of the families we happened to examine.

Author contributions

A.M. conceived the study, developed the theory and its evaluation, implemented the computational work, designed the validation, edited the manuscript, oversaw all agentic workflows, and funded the project personally. Z.G. directed the scope of the work, independently checked and pressure-tested the derivations, identified the estimator-projection issue that led to the revision of the paper's central claim, and reviewed the manuscript throughout. Both authors take responsibility for the work.

Use of AI tools. An AI assistant (Anthropic's Claude models, through the Claude Code command-line interface, 2026 releases) was used for the initial derivations of Section 3, the implementation and its test suite, the simulations and their analysis, the figures, the initial literature and bibliography search, the drafting of the text, and adversarial attempts to falsify the claims of Section 3. The authors set the research questions, chose the definitions and the validation design, verified every derivation, and selected what to report; no result is asserted on the assistant's word.

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The coupling map behind the transpilation counts of Section 2.6, and the device measurements committed with the repository but not analysed here, were obtained through public-tier access to the Open Quantum platform, whose provision of that access we acknowledge.

Methods, data, and code

Every number in this paper is produced by a script in the public repository at `https://github.com/alex-messerlian/shadow-moment-crossover`, which also holds the generated artifacts each figure and table is built from.

The computational work has three layers. The exact evaluators are sampling-free: the statewise projection variances of Section 3.5 are contractions of the second-moment identity computed in exact or double-precision arithmetic, with the ensemble-averaged closed forms carried in exact rational arithmetic so no size overflows. The estimators of Section 5 and the crossover sweeps are Monte Carlo over simulated local-shadow snapshots, with all randomness drawn from seeded, value-based generators, so every figure regenerates bit-for-bit from the committed seeds. The verification layer is a test suite that locks each derivation as an executable check: the identities against independent brute-force computations, the closed forms against their Haar averages, the estimators against their exact targets, and the chunked implementations against their unchunked values.

Simulations used Python 3.12 with NumPy 2.5, SciPy 1.18 and Matplotlib 3.11; the transpiled gate counts of Section 2.6 used Qiskit 2.5 against a coupling map carried as committed package data, so they reproduce offline with no credentials. Sizes reported as feasible were timed on a single core.

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A Exact evaluations of the fourth-moment U-statistic and the two projection variances

The full U-statistic for $\mathrm{Tr}(\rho^4)$ requires the sum over all distinct ordered 4-tuples of snapshots,

$$\mathcal{S} = \sum_{(i,j,k,l)\ \text{pairwise distinct}} \mathrm{Tr}(G_i G_j G_k G_l).$$

Because the snapshots do not commute, this does not reduce to matrix power sums. It is obtained by Möbius inversion over the partition lattice of the four cyclic slots.

For each set partition P of $\{0,1,2,3\}$, let $T(P)$ be the sum of $\mathrm{Tr}(G_{a_0} G_{a_1} G_{a_2} G_{a_3})$ over all index assignments in which slots in the same block share an index and different blocks are unconstrained. The Möbius function of the partition lattice is

$$\mu(P) = \prod_{b\in P} (-1)^{|b|-1}(|b|-1)!,$$

and the all-distinct sum is

$$\mathcal{S} = \sum_{P} \mu(P)\, T(P),$$

summed over the 15 set partitions of four elements.

Most $T(P)$ reduce to matrix power sums built from $S_1 = \sum_i G_i$, $S_2 = \sum_i G_i^2$, $S_3 = \sum_i G_i^3$ and $S_4 = \sum_i G_i^4$, kept distinct from the set partitions P and from the purity $p_2 = \mathrm{Tr}(\rho^2)$ of the main text. The exception is the alternating partition $\{0,2\}\{1,3\}$, which requires

$$A = \sum_{i,j} \mathrm{Tr}(G_i G_j G_i G_j),$$

and this has no power-sum representation. It is computed exactly by a tensor contraction. Define the $d^2 \times d^2$ matrix

$$X_{(a,b),(c,d)} = \sum_i (G_i)_{ab}(G_i)_{cd} = \sum_i \mathrm{vec}(G_i)\, \mathrm{vec}(G_i)^\top,$$

reshape it to X_{abcd}, and contract

$$A = \sum_{a,b,c,d} X_{abcd}\, X_{bcda}.$$

Forming X costs $\Theta(Md^4)$ in time, since each of the M rank-one updates touches all d^4 entries, and $\Theta(d^4)$ in memory; the two contractions cost $\Theta(d^4)$. The construction as written is therefore $\Theta(Md^4)$. The estimator used for the results reported here never forms X: it evaluates the alternating term through the per-qubit factorization $\mathrm{Tr}(G_i G_j G_i G_j) = \prod_q \mathrm{Tr}(G_i^{(q)} G_j^{(q)} G_i^{(q)} G_j^{(q)})$ as an $M \times M$ contraction, which costs $O(M^2 n + Mnd^2)$ and forms no d^4 tensor. Both routes agree, and the complete $k = 4$ estimator matches brute-force enumeration over all distinct 4-tuples to $\lesssim 10^{-13}$.

The two-copy spectral sum. The $k = 2$ kernel $\mathrm{Tr}(G_1 G_2)^2$ is diagonal in the Pauli basis with per-qubit weights $\{I : 112,\ X, Y, Z : 8\}$, so its expectation over a state with Pauli expansion $\rho = 2^{-n} \sum_P \langle P \rangle P$ is $7^n \sum_P \langle P\rangle^2 (1/14)^{|P|}$: each qubit multiplies the identity slot by $112/16 = 7$ and every non-identity slot by $8/16 = 1/2 = 7 \cdot (1/14)$. Subtracting $\mathrm{Tr}(\rho^2)^2$ for the disconnected part gives the spectral identity of Section 3.5. At the maximally mixed state only the identity term survives, returning $7^n - 4^{-n}$; on the noisy-pure ensemble the weighted counts 28^n and 34^n of the main text follow. This is verified against exact enumeration to machine precision by `experiments/beta_law_regenerate.py`.

The first projection variance. The same second-moment identity fixes $\bar{\zeta}_1$, with the counting run over pairs of strings rather than over single strings. Writing a snapshot in the Pauli basis as $G = 2^{-n} \sum_P \hat{x}_P P$, the first-order projection is $\mathbb{E}_{G'}[\mathrm{Tr}(GG')] = \mathrm{Tr}(G\rho) = 2^{-n} \sum_P \hat{x}_P \langle P \rangle$, so

$$\zeta_1 = 4^{-n} \sum_{P,Q} \langle P \rangle \langle Q \rangle\, \mathbb{E}[\hat{x}_P \hat{x}_Q] \;-\; \mathrm{Tr}(\rho^2)^2.$$

Under local shadows a qubit measured in one basis says nothing about another, so $\mathbb{E}[\hat{x}_P \hat{x}_Q]$ vanishes unless P and Q are *compatible*: on every qubit either $P_i = Q_i$ or one of the two is the identity. For a compatible pair the second-moment identity gives $\mathbb{E}[\hat{x}_P \hat{x}_Q] = 3^a \langle P \ominus Q \rangle$, where $a = \#\{i : P_i = Q_i \neq I\}$ counts the shared support and $P \ominus Q$ is the string supported where exactly one of the two is non-identity, the same string as in Section 3.5.

Haar-averaging then splits the pair sum into four blocks, according to how many of P, Q and $P \ominus Q$ are the identity. The both-identity block contributes 1. The diagonal $P = Q \neq I$ contributes $3^{|P|} \langle P \rangle^2$, and $\sum_P 3^{|P|} = 10^n$ because each qubit offers $1 + 3 \cdot 3$; the second Haar moment sends each $\langle P \rangle^2$ to $u^2/(d+1)$. The two strips with one argument the identity contribute $\langle Q \rangle^2$ apiece over the $4^n - 1$ non-identity strings. Only the last block, $P \neq Q$ with both non-identity, needs the third moment: there P, Q and $P \ominus Q$ are all non-identity and satisfy $PQ \propto P \ominus Q$. Its weight is the compatible-pair total $\sum 3^a = 16^n$, each qubit offering $1+3+3+3\cdot 3$, less the three blocks already counted. Assembling the four blocks gives the closed form of Section 3.5. Direct enumeration of all 16^n pairs reproduces the four counts, and the assembled $\bar{\zeta}_1$ and $\bar{\zeta}_2$ agree with `closed_form_zetas` exactly in rational arithmetic for $n = 2$ to 5.

B The destructive SWAP test

Two copies of an n-qubit state occupy qubits $0 \ldots n-1$ (copy A) and $n \ldots 2n-1$ (copy B). For each $i \in \{0, \ldots, n-1\}$, apply a CNOT with control i and target $i+n$, then a Hadamard on qubit i. Measure all $2n$ qubits. For each measured bitstring, let a_i and b_i be the outcomes on qubits i and $i+n$. The purity is

$$\mathrm{Tr}(\rho^2) = \sum_{\text{bitstrings}} (-1)^{\#\{i\,:\,a_i=b_i=1\}}\, P(\text{bitstring}).$$

The circuit uses exactly n two-qubit gates, no ancilla, and has depth two after state preparation. The sign rule is invariant under bitstring reversal, so the endianness convention of the backend does not affect the result. We verified the rule against exact simulation for pure and mixed states to 10^{-16}.

C Reproducibility and statistics

This appendix records the sampling design behind every headline number, so that the agreement between prediction and measurement can be judged against how the two were generated. All counts are read from the committed generating code; the sizes and seeds below are those actually used.

State and snapshot counts. The projection variances ζ_c that feed the theory curves are estimated from 4 independently drawn states per (n, k), each from 120,000 shadow snapshots (`results/theory_zetas.json`: `n_states` = 4, `n_samples` = 120,000). At $k = 2$ these coefficients are now computed in closed form (Section 3.5) and carry no sampling error; the sampled values are retained in the repository, and substituting them changes no crossover prediction and no α verdict. The measured α points and the noisy-pure crossover cells are averaged over 48 independently drawn states per size, with 8 Monte Carlo trials each in the 60 cells of `budget_scaling.json` (`run_budget_sweep.py`: `N_STATES` = 48, `N_TRIALS` = 8) and 10 in the 36 cells of `moment_sweep_corrected.json`. The held-out stress ensembles use 4 states for the variance estimate and 24 for the measurement on the random families (Haar-pure, low-rank), and a single fixed state with 36 trials for the deterministic GHZ-noisy family (`run_stress_test.py`); their variance estimates use 60,000 snapshots. All states in these sweeps are depolarized Haar-random pure states at $q = 0.1$, except the Haar-pure, low-rank and GHZ-noisy families defined in Section 6.3; the two varying-parameter families are defined in Section 4.1. The runs that use them carry their own designs. The ordering result of Section 4.2 draws 14 states per cell on variable-rank at $n = 3, 4, 5$ and budgets 2000 and 8000, with 300 forward-simulation trials per state, and repeats the same run on noisy-pure as the control (`seed` = 47); the five-family crossover sweep of Section 4.3 is a separate run, whose prediction side is the sampling-free statewise evaluator rather than the 4 sampled states above; it uses 24 prediction states and 24 measurement states per cell at 6 trials each, over sizes 2 to 6, three channels and rates 0.05, 0.1 and 0.3 at budget 2000 (`seed` = 0), with the deterministic GHZ-noisy family again a single state at 36 trials.

The estimand is degenerate on three of the four ensembles. For $\rho = (1-q)|\psi\rangle\langle\psi| + qI/d$ the spectrum is $(1-q) + q/d$ once and q/d with multiplicity $d-1$ whatever $|\psi\rangle$ is, so $\mathrm{Tr}(\rho^k) = \big((1-q)+q/d\big)^k + (d-1)(q/d)^k$ depends on (n, q, k) alone; the same holds trivially for Haar-pure states, whose moments are all 1, and for the single fixed GHZ-noisy state. Over 48 states at $n = 4$ the per-state standard deviation of $\mathrm{Tr}(\rho^k)$ is at machine precision on all three, at most 7.1×10^{-16} (noisy-pure), 1.2×10^{-15} (Haar-pure), 1.1×10^{-16} (GHZ-noisy), over $k = 2, 3, 4$, while only the rank-2 Ginibre family varies, at 5.0%, 12.9%, and 21.9% relative spread.

Averaging over 48 states therefore averages estimator *difficulty* at a fixed estimand, not distinct estimation problems; that weaker reading is the one we mean. The difficulty does vary: the per-state ζ_2 has relative spread 1.6% at $n = 4$ on noisy-pure, being a weighted sum over the state's Pauli spectrum (Section 3.5) rather than a function of ρ's spectrum. The estimator is never given q or the ensemble. What the degeneracy costs is robustness testing against a varying target, which the rank-2 family supplies too weakly to resolve an ordering and the varying-parameter families of Section 4.1 supply properly.

Train/test separation. The prediction and the measurement it is compared against do not share snapshots. The ζ_c are estimated from one pseudorandom substream (seeded $[\,0, n, s, k, 3\,]$) and the measured RMSE from a disjoint one (seeded $[\,0, n, s, 1, k, t\,]$), so the shot noise in the theory curve and the shot noise in the measured points are independent by construction. The theory curve is never fit to the measured α: it is the exact-variance RMSE evaluated at the estimated ζ_c and fit over the budget grid on its own (`figures/data.py`: the components are read from `theory_zetas.json` and "not re-estimated"). The state draws do overlap: the four states used for ζ_c are, at this seed, the first four of the forty-eight used for the measured points, since both draw ρ from $\texttt{noisy\_pure}(n, 0.1, \mathrm{rng}[0, n, s, 0])$. The agreement is therefore not built in through shared shots, though it is not a fully held-out state set either.

Bootstrap resampling units. The α standard errors resample the state as the independent unit (2000 resamples, each carrying its whole correlated budget vector; `fit_budget_exponent_bootstrap`), which is the honest unit because the nested budgets share a draw.

The α regression. Both the measured and the theory α are the negative ordinary-least-squares slope of log RMSE against $\log M$ over the same per-cell budget grid ($k = 2$: 2000–128,000, capped to 2000–32,000 for $n \geq 7$; $k = 3$: 2000–32,000; $k = 4$: 500–8000), with $\mathrm{RMSE}(M) = \sqrt{\mathrm{MSE}}$ averaged over states. The two differ only in their input, measured MSE versus the exact-variance RMSE, and are otherwise the identical procedure.

Crossover definition. The crossover n^* is an integer drawn from the swept sizes; no interpolation is used, and a cell whose curves do not cross inside the tested range is censored (returns none) and excluded from the aggregate rather than counted as a hit or a miss. A crossing is *sustained* when the collective route wins at n^* and at every larger swept size, so a single crossing that later reverses does not count. Three rules produce the numbers reported here, and they are not interchangeable:

	compares	verdict	produces
1	theory against theory	sustained	every *predicted* n^*
2	measured against measured	state-level paired z-test, first win at $\lvert z\rvert > 2$, sustained or not	the measured n^* of the 68 noisy-pure cells
3	measured single-copy RMSE against the theory collective floor	sustained	the measured n^* of the held-out cells, 15 of which enter the 83

Rule 1 is `predict_crossover` (`theory/crossover.py`), rule 2 `crossover_table` (`benchmark/hardened.py`), rule 3 part 4 of `run_stress_test.py`.

The rules agree on 88 of the 96 noisy-pure cells swept. In seven of the eight disagreements the paired test is the more conservative, declining a call the sustained rule takes on a margin indistinguishable from zero ($|z|$ between 0.02 and 1.47; single-copy RMSE at most 1.06 times collective) and taking the crossover one qubit later, where $|z|$ runs 3.7 to 12.0. The eighth is flagged `ambiguous` by the pipeline: the collective route wins at $n = 2$ but the two tie at $n = 5$, so no sustained crossing exists.

Fit uncertainty. The exponential-fit residuals in Section 3.7 are positive at both ends of the range ($+0.20$ at $n = 2$, $+0.13$ at $n = 9$ in $\log M^*$) and negative through the middle, the signature of mild upward curvature, and that is why the fitted base rises with the window as Section 3.5 reports. Refitting with each n dropped in turn moves the base within $[5.25, 5.49]$, with the low-n point exerting the most leverage (dropping $n = 2$ raises it to 5.49). The base is a finite-size fit over the sizes reached, not the asymptotic rate, which Section 3.5 gives in closed form.

The full crossover table. The 83 resolved cells break down as: 68 noisy-pure cells drawn from two independent sweeps (`budget_scaling.json` and `moment_sweep_corrected.json`), which comprise 54 distinct (k, channel, rate, budget) configurations with 14 evaluated in both sweeps (and all

14 replications agree on both the measured and the predicted n^*) plus 15 cells across the three held-out ensembles. Of the 123 cells swept in total, these 83 resolved a sustained crossover within the tested range; the remaining 40 (28 noisy-pure and 12 held-out) were censored because their single-copy and collective curves do not cross inside the swept range, and are excluded from the aggregate rather than scored as a hit or a miss. Treating "no crossover in the swept range" as a valid prediction and scoring all swept cells, the criterion is exact on 109 and within one qubit on 118, or 88.6% and 95.9%; the censored cells are not random, concentrating at the largest budgets, where the crossover is pushed past the swept range, and in the Haar-pure ensemble. Among the censored cells the criterion agrees on no crossover in 36 and misses in 4, so the censoring discards mostly correct no-crossover predictions but also removes four misses that would lower the within-one-qubit figure to 95.9%. The supplementary material lists all 83 individually, and the same cells are committed as `budget_scaling.json`, `moment_sweep_corrected.json`, and `stress_test.json` in the repository. One cell misses by more than one qubit (noisy-pure, $k = 4$, predicted 4 against measured 2), giving $82/83 = 98.8\%$ within one qubit and $73/83 = 88.0\%$ exact. Stratified by the measured crossover, all 52 cells with $n^*_{\text{meas}} \geq 3$ fall within one qubit and 43 are exact; the one miss beyond a qubit sits in the 31-cell $n^*_{\text{meas}} = 2$ group. Counted as configurations rather than evaluations, they are 69 independent settings.